

quantum optimization and machine learning

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Chapter 0

Introduction

0.1 Classical vs Quantum Bits

Classical bits

- can be in two distinct states, 0 and 1
- can be measured completely
- are not changed by measurement
- can be copied/erased

Quantum bits

- can be in state $|0\rangle$ or $|1\rangle$ or in any other state that is a linear combination of the two.
- ...
- cannot be copied/erased

Advantages of qubits:

- adding qubits increases storage exponentially
- can perform operations on all superpositions

0.2 Properties

Superposition:

$$|\Psi\rangle = \alpha_0|0\rangle + \alpha_1|1\rangle, |\alpha_0|^2 + |\alpha_1|^2 = 1 \quad (1)$$

Entanglement:

- When a pair or group of particles is generated, interact, or share spatial proximity in a way such that the quantum state of each particle of the pair or group cannot

be described independently of the state of the others, even when the particles are separated by large distance.

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Decoherence:

- Q. dec. is the loss of superposition because of a spontaneous interaction between a quantum system and its environment.
- One of the reasons why qc have a long way to go bc. superposition and entanglement are extremely fragile states.

0.3 Quantum Computing

Programs are executed by unitary evolution of an input that is given by the state of the system. Any program is modeled as a unitary transformation! (Note, because all unitary operators are invertible, qc is always reversible!)

Measurement:

- quantum measurement is a decoherence process (if the system were isolated, it would maintain coherence indefinitely, but it would be impossible to manipulate or investigate it).
- When a system is measured, the wave function collapses to a new state according to a probabilistic queue.

0.4 The Status Quo

NISQ

- Noisy Intermediate Stage Quantum Computing
- parametric Quantum Circuit Learning (hybrid approach between quantum and classical computing)

0.5 Course Outline

- Foundations of quantum computing: qubits, quantum circuits, entanglement
- Quantum algorithms: gate-based fully quantum algorithms - quantum advantage
- Quantum optimization: quantum annealing and hybrid (NISQ) algorithms
- Quantum machine learning: hybrid (NISQ) algorithms

Chapter 1

1.1 Single qubit

1.1.1 Dirac's notation

- Column vector: ket

$$|x\rangle = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{C}^n$$

- Its conjugate transpose: bra

$$\langle x| = [x_1^*, \dots, x_n^*] \in \mathbb{C}^n$$

- Inner product: $\langle x|y\rangle = \sum_{i=1}^n x_i^* y_i$
- Norm: $\|x\| = \sqrt{\langle x|x\rangle}$
- Outer product: $|x\rangle\langle y|$ = rank-1 matrix
- Hilbert space $\mathcal{H}$: inner product vector space
- Orthonormal basis: $B = \{|b_0\rangle, \dots, |b_{n-1}\rangle\}$ such that $\langle b_i|b_j\rangle = 1$ if $i = j$ and 0 if $i \neq j$. For any vector in this space we can write it as $|x\rangle = \sum_{i=0}^{n-1} a_i |b_i\rangle$ and the norm of its projection $a_i = \langle b_i|x\rangle$.

1.1.2 Quantum state - qubit

Quantum computing involves three steps:

1. Start by preparing some state given some classical data (qubits)
2. Perform quantum operations (unitary transformation)
3. perform measurement (returns classical data)

Postulate 1

For any isolated system, at any time, it is completely specified by a state $|\Psi\rangle \in \mathcal{H}$ with $\|\Psi\| = 1$.

Each qubit is in $\mathcal{H} = \mathbb{C}^2$

The computational basis is $|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $|1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$. This means that each qubit can be written as

$$|\Psi\rangle = \alpha|0\rangle + \beta|1\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}, \quad \alpha, \beta \in \mathbb{C} \text{ and } |\alpha|^2 + |\beta|^2 = 1.$$

Two qubits are equivalent if they differ by a global phase. For example:

$$\begin{aligned} |\Psi_0\rangle &= |\Psi\rangle \\ |\Psi_1\rangle &= e^{j\theta}|\Psi\rangle \end{aligned}$$

Bloch sphere representation of a qubit

$$\begin{aligned} \alpha &= r_0 e^{j\theta_0} \text{ and } \beta = r_1 e^{j\theta_1}, r_0^2 + r_1^2 = 1 \\ \Rightarrow |\Psi\rangle &= r_0 e^{j\theta_0} |0\rangle + r_1 e^{j\theta_1} |1\rangle = r_0 |0\rangle + r_1 e^{j(\theta_1 - \theta_0)} |1\rangle = \cos(\phi/2) |0\rangle + \sin(\theta/2) e^{j\phi} |1\rangle \end{aligned}$$

Where $0 \leq \theta \leq \pi$ and $0 \leq \phi \leq 2\pi$.

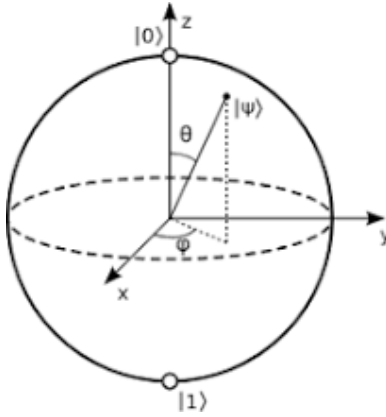


Figure 1.1: Bloch sphere

The axes on the Bloch sphere are defined by the eigenvectors of each axis operator.

Other vectors to know:

- $|+\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$
- $|-\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

- If $\theta = 0$, then $|\Psi\rangle = |0\rangle$.
- If $\theta = \pi$, then $|\Psi\rangle = |1\rangle$.

Therefore:

- The z-axis is the computational basis
- The x-axis is the Hadamard basis
- The y-axis another basis...

1.1.3 Quantum Operations - Unitary transformations

Postulate 2

A closed system evolves according to a unitary operator (for this course, an operator is just a square matrix). For any input $|\Psi\rangle$, the output of an operator is just $U|\Psi\rangle$.

Recall: A matrix U is unitary if it is $U^*U = UU^* = I$ and $U^{-1} = U^*$.

Properties:

- eigenvalues of U have the unit norm $\lambda_k = e^{j\theta_k}$
- U preserves inner product and norm (otherwise it violates Postulate 1)

Qubit operators - quantum gates

If $U|0\rangle = |u_1\rangle$ and $U|1\rangle = |u_2\rangle$ then we can write $U \begin{bmatrix} |0\rangle & |1\rangle \end{bmatrix} = \begin{bmatrix} |u_1\rangle & |u_2\rangle \end{bmatrix} = UI = U$.

- X-gate (bit flip):

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

- Z-gate (phase flip):

$$\begin{cases} Z|0\rangle = |0\rangle \\ Z|1\rangle = -|1\rangle \end{cases} \Rightarrow Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

- H-gate (Hadamard operator):

$$\begin{cases} H|0\rangle = |+\rangle \\ H|1\rangle = |-\rangle \end{cases} \Rightarrow H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Pauli operators I, X, Y, Z

$$\begin{cases} Y|0\rangle = j|1\rangle \\ Y|1\rangle = -j|0\rangle \end{cases} \Rightarrow Y = \begin{bmatrix} 0 & -j \\ j & 0 \end{bmatrix}$$

Eigendecomposition:

$$\begin{aligned}
I &= |v_0\rangle\langle v_0| + |v_1\rangle\langle v_1|, \quad \{|v_0\rangle, |v_1\rangle\} \text{ any orthonormal vectors} \\
X &= |+\rangle\langle +| - |-\rangle\langle -| \\
Z &= |0\rangle\langle 0| - |1\rangle\langle 1| \\
Y &= |+j\rangle\langle +j| - |-j\rangle\langle -j|, \quad |+\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ j \end{bmatrix}, |-\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -j \end{bmatrix}
\end{aligned}$$

Properties:

- Cyclic product property:
 - $XY = jZ$
 - $YZ = jX$
 - $ZX = jY$
- Anti-commuting property:
 - $XY = -YX$
 - $YZ = -ZY$
 - $ZX = -XZ$

Pauli rotational operators

Recall: Any unitary operator U can be expressed as $U = \exp(-jG)$ where G is Hermitian ($G^* = G$).

Recall: If a matrix G is Hermitian then it has a spectral decomposition with all real eigenvalues:

$$G = \sum_i \lambda_i |\lambda_i\rangle\langle \lambda_i|$$

Therefore a function of a matrix is simply

$$f(G) = \sum_i f(\lambda_i) |\lambda_i\rangle\langle \lambda_i|$$

Finally this allows us to write:

$$U = \exp(-jG) = \sum_i e^{-j\lambda_i} |\lambda_i\rangle\langle \lambda_i|$$

The eigenvalue $e^{-j\lambda_i}$ has unit norm!

The Pauli operators X, Y, Z are Hermitian for $Q \in \{X, Y, Z\}$, so the rotation operators are

$$\begin{aligned}
R_Q(\theta) &= \exp\left(-j\frac{\theta}{2}Q\right) \\
&= \exp\left(-j\frac{\theta}{2}\right)|v_0\rangle\langle v_0| + \exp\left(j\frac{\theta}{2}\right)|v_1\rangle\langle v_1| \\
&= \cos\left(\frac{\theta}{2}\right)[|v_0\rangle\langle v_0| + |v_1\rangle\langle v_1|] - j\sin\left(\frac{\theta}{2}\right)[|v_0\rangle\langle v_0| - |v_1\rangle\langle v_1|] \\
&= \cos(\theta/2)I - j\sin(\theta/2)Q
\end{aligned}$$

Special cases:

$$\begin{aligned}
R_Q(\pi) &= Q, \quad Q \in \{X, Y, Z\} \\
R_x(\theta) &= \begin{bmatrix} \cos(\pi/2) & -j\sin(\pi/2) \\ -j\sin(\theta/2) & \cos(\theta/2) \end{bmatrix} \\
R_y(\theta) &= \begin{bmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{bmatrix} \\
R_Z(\pi/2) &= S = \begin{bmatrix} 1 & 0 \\ 0 & e^{j\pi/2} \end{bmatrix} \\
R_Z(\pi/4) &= T = \begin{bmatrix} 1 & 0 \\ 0 & e^{j\pi/4} \end{bmatrix} \\
T^2 &= S, S^2 = Z \\
R_Z(\theta) &= \begin{bmatrix} e^{-j\theta/2} & 0 \\ 0 & e^{j\theta/2} \end{bmatrix} \equiv \begin{bmatrix} 1 & 0 \\ 0 & e^{j\theta} \end{bmatrix} = P(\text{definition of } P)
\end{aligned}$$

1.1.4 Quantum measurement

Postulate 3 - Von Neumann measurement

Given orthonormal basis $\{|b_0\rangle, \dots, |b_{n-1}\rangle\}$. The system state is $|\Psi\rangle$ before measurement. The measurement outcome is $(m = i)$ with probability $|\langle b_i | \Psi \rangle|^2$.

If the outcome is $m = i$, then the post-measurement state is $|\Psi_{post}^{(i)}\rangle = |b_i\rangle$.

Qubit example: $|\Psi\rangle = \alpha|0\rangle + \beta|1\rangle$

1. If measured using computational basis $|0\rangle, |1\rangle$, we will see

- “0” w.p. $|\langle 0 | \Psi \rangle|^2 = |\alpha|^2$
- “1” w.p. $|\langle 1 | \Psi \rangle|^2 = |\beta|^2$

Therefore, α and β are known as probability amplitudes

2. If measured using the Hadamard basis $|+\rangle, |-\rangle$, we will see

- “+” w.p. $|\langle +|\Psi\rangle|^2 = |\langle +|(\alpha|0\rangle + \beta|1\rangle)|^2 = |\alpha\langle +|0\rangle + \beta\langle +|1\rangle|^2 = |\frac{\alpha}{\sqrt{2}} + \frac{\beta}{\sqrt{2}}|^2 = \frac{|\alpha+\beta|^2}{2}$.
- “-” w.p. $\frac{|\alpha-\beta|^2}{2}$

Note: For quantum computing, we always measure with respect to the computational basis (aka standard measurement).

If we want to measure with respect to a non-computational basis:

- Any unitary operator $U = [|u_0\rangle \quad |u_1\rangle] \cdot I = |u_0\rangle\langle 0| + |u_1\rangle\langle 1|$ (changed from computational to U -basis)
- If you have a state, you can convert to a new basis by just multiplying it by a unitary operator. $U = |u_0\rangle\langle 0| + |u_1\rangle\langle 1|$
- Similarly, $U^* = |0\rangle\langle u_0| + |1\rangle\langle u_1|$ changes from U -basis to computational basis.

1.2 Multiple Qubits

1.2.0 Tensor Product

For vectors:

$$|x\rangle = \begin{bmatrix} x_1 \\ \vdots \\ x_m \end{bmatrix} \in \mathbb{C}^{m \times 1}, |y\rangle = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} \in \mathbb{C}^{n \times 1} \implies |x\rangle \otimes |y\rangle = \begin{bmatrix} x_1|y\rangle \\ \vdots \\ x_n|y\rangle \end{bmatrix} \in \mathbb{C}^{mn \times 1}$$

Similarly, for matrices $A \in \mathbb{C}^{m \times n}$ and $B \in \mathbb{C}^{p \times q}$:

$$A \otimes B = \begin{bmatrix} a_{11}B & \dots & a_{1n}B \\ \vdots & \ddots & \vdots \\ a_{m1}B & \dots & a_{mn}B \end{bmatrix} \in \mathbb{C}^{mp \times nq}$$

Properties

- $(A \otimes B)^\top = A^\top \otimes B^\top$
- $(A \otimes B)^* = A^* \otimes B^*$
- $(A \otimes B)^{-1} = A^{-1} \otimes B^{-1}$
- mixed-product law: $(A \otimes B)(C \otimes D) = AC \otimes BD$
 - A, B are Hermitian, then $A \otimes B$ is Hermitian
 - A, B are unitary, then $A \otimes B$ is unitary
- No change of order unlike inverse and conjugate transpose on general matrices

Tensor Product Space

Given two Hilbert spaces H_1 and H_2 with bases $\{|b_i\rangle, i = 1, \dots, d_1\} \rightarrow H_1$ and $\{|c_j\rangle, j = 1, \dots, d_2\} \rightarrow H_2$ where $d_1 = \dim(H_1)$ and $d_2 = \dim(H_2)$:

Then their tensor product space is:

$$H = H_1 \otimes H_2 = \left\{ \sum_i a_i |x_i\rangle \otimes |y_i\rangle, \quad \text{where } |x_i\rangle \in H_1, |y_i\rangle \in H_2, a_i \in \mathbb{C} \right\}$$

$$\Rightarrow \dim(H) = d_1 \cdot d_2$$

$$\Rightarrow \text{basis of } H : \{|b_i\rangle \otimes |c_j\rangle, i = 1, \dots, d_1, j = 1, \dots, d_2\}$$

Inner product on H :

$$(\langle x-1 | \otimes \langle y_1 |)(|x_2\rangle \otimes |y_2\rangle) = \langle x_1|x_2\rangle \langle y_1|y_2\rangle$$

$$(A \otimes B)|x\rangle \otimes |y\rangle = A|x\rangle \otimes B|y\rangle$$

1.2.1 Multi-qubit state

Postulate 4

A system composed of two subsystems H_1 and H_2 must be treated as $H = H_1 \otimes H_2$.

- If $|\Psi_1\rangle \in H_1$ and $|\Psi_2\rangle \in H_2 \Rightarrow |\Psi_1\rangle \otimes |\Psi_2\rangle \in H$.

Separable State

If $|\Psi\rangle \in H = H_1 \otimes H_2$ can be written as $|\Psi\rangle = |\Psi_1\rangle \otimes |\Psi_2\rangle$ where $|\Psi_1\rangle \in H_1$ and $|\Psi_2\rangle \in H_2$ then $|\Psi\rangle$ is separable.

- This means that $(U_1 \otimes U_2)(|\Psi_1\rangle \otimes |\Psi_2\rangle) = (U_1|\Psi_1\rangle) \otimes (U_2|\Psi_2\rangle)$.
- If one of the states like $U_2 = I$, then this becomes $U_1|\Psi_1\rangle \otimes |\Psi_2\rangle$ (allows us to operate on only H_1).
- In general it is not possible to do this!
- Otherwise, if a state is not separable, it is entangled!

2-qubit system

$H_1 = H_2 = \mathbb{C}^2$. the basis of H_1, H_2 is $|0\rangle, |1\rangle$.

Therefore, the basis of $H = H_1 \otimes H_2$ is (in different notations):

$$[|0\rangle \otimes |0\rangle, |0\rangle \otimes |1\rangle, |1\rangle \otimes |0\rangle, |1\rangle \otimes |1\rangle] = [|00\rangle, |01\rangle, |10\rangle, |11\rangle] = [|0\rangle_2, |1\rangle_2, |2\rangle_2, |3\rangle_2] = I_4$$

In general, a 2-qubit state $|\Psi\rangle \in H$ can be written as

$$|\Psi\rangle = a_{00}|00\rangle + a_{01}|01\rangle + a_{10}|10\rangle + a_{11}|11\rangle, \text{ s.t. } |a_{00}|^2 + |a_{01}|^2 + |a_{10}|^2 + |a_{11}|^2 = 1$$

This cannot always be written as $|\Psi_1\rangle \otimes |\Psi_2\rangle$!!

1.2.2 Multi-qubit gates

For a multi qubit gate with n inputs, the top qubit is always indexed 0 (in Qiskit) and is the least significant bit:

$$|q_{n-1}\rangle \otimes \dots \otimes |q_1\rangle \otimes |q_0\rangle$$

These operations are still unitary, but have dimension $2^n \times 2^n$.

$$U = [|u_0\rangle, |u_1\rangle, |u_2\rangle, |u_3\rangle] \in \mathbb{C}^{4 \times 4}$$

Separable operators are ones that can be written as $U = U_1 \otimes U_2$ where $U_1, U_2 \in \mathbb{C}^{2 \times 2}$

A gate U is simply a change of basis:

$$U|\Psi\rangle = U \sum_{k=0}^k a_k |k\rangle = \sum_{k=0}^3 a_k \cdot U|k\rangle$$

If both U and $|\Psi\rangle$ are separable, you can apply U as such:

$$U|\Psi\rangle = (U_1 \otimes U_2)(|\Psi_1\rangle \otimes |\Psi_2\rangle) = U_1|\Psi_1\rangle \otimes U_2|\Psi_2\rangle$$

1. $H^{\otimes n}$ gate:

$$\begin{aligned} H &= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \\ H^{\otimes 2} &= H \otimes H = \frac{1}{\sqrt{2}} \begin{bmatrix} H & H \\ H & -H \end{bmatrix} \\ \Rightarrow H^{\otimes n} &= \frac{1}{\sqrt{2}} \begin{bmatrix} H^{\otimes n-1} & H^{\otimes n-1} \\ H^{\otimes n-1} & -H^{\otimes n-1} \end{bmatrix} \end{aligned}$$

Property: balanced superposition of all computational basis

$$H^{\otimes n}|0\rangle_n = \frac{1}{\sqrt{2^n}} \sum_{i=0}^{2^n-1} |i\rangle_n$$

2. Controlled- U gate:

- if control bit is $|1\rangle$ then apply U to target bit (otherwise do not do anything)

If we have CU (with control q_0 and target q_1):

- $CU|00\rangle = |00\rangle$

$$\bullet \quad CU|01\rangle = U|0\rangle \otimes |1\rangle = \begin{bmatrix} u_{00} \\ u_{10} \end{bmatrix} \otimes \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ u_{00} \\ 0 \\ u_{10} \end{bmatrix}$$

- $CU|10\rangle = |10\rangle$
- $CU|11\rangle = U|1\rangle \otimes |1\rangle = \begin{bmatrix} u_{01} \\ u_{11} \end{bmatrix} \otimes \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ u_{01} \\ 0 \\ u_{11} \end{bmatrix}$

Therefore:

$$CU = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & u_{00} & 0 & u_{01} \\ 0 & 0 & 1 & 0 \\ 0 & u_{10} & 0 & u_{11} \end{bmatrix}$$

"This is how nature works" - XW

If we flip it with control q_1 and target q_0 :

- $CU|00\rangle = |00\rangle$
- $CU|01\rangle = |01\rangle$
- $CU|10\rangle = |1\rangle \otimes U|0\rangle$
- $CU|11\rangle = |1\rangle \otimes U|1\rangle$

This gives us $CU = \begin{bmatrix} I & 0 \\ 0 & U \end{bmatrix}$

Important controlled gates:

- CX or CNOT
- CZ (can be just a line with two dots because it doesn't matter which one is the control or target!)
 - $|00\rangle \rightarrow |00\rangle$
 - $|01\rangle \rightarrow |01\rangle$
 - $|10\rangle \rightarrow |10\rangle$
 - $|11\rangle \rightarrow |(-1)1\rangle = -|11\rangle$

3. SWAP gate: denoted by a line with two x 's on each qubit

- $|00\rangle \rightarrow |00\rangle$
- $|01\rangle \rightarrow |10\rangle$
- $|10\rangle \rightarrow |01\rangle$
- $|11\rangle \rightarrow |11\rangle$

This does not switch the qubits but instead changes the basis vectors!!

Example: $H^{\otimes 2}CXH^{\otimes 2}|\Psi\rangle \equiv rCX$ reverse cnot!

Universal gates

The set of gates $\{CX, H, T\}$ is universal, meaning that you can construct any unitary gate using these three gates.

No-cloning theorem

There is no universal copier of quantum state - U such that the following is true:

$$U|\Psi\rangle \otimes |\Phi\rangle = |\Psi\rangle \otimes |\Psi\rangle$$

This would violate both linearity and unitarity (if you copy and overwrite the information, you can no longer invert this operation and get the information back).

1.2.3 Multi-qubit measurement

Von Neumann measurement (complete measurement)

Orthonormal basis $\{|u_0\rangle, |u_1\rangle, \dots, |u_{N-1}\rangle$ where $N = 2^n$

Output $m = \{0, \dots, N-1\}$, with $P(m=x) = |\langle u_x|\Psi\rangle|^2 = \langle \Psi|u_x\rangle\langle u_x|\Psi\rangle$.

We notice that the $|u_x\rangle\langle u_x|$ is a rank-1 matrix (projector).

Post measurement state $|\Psi_{\text{post}}\rangle = |u_x\rangle$.

Again, we typically measure in the computational basis, but for general measurement we first apply U^* , measure, then apply U again.

Take 2: Projective measurement

A projector is Hermitian $P^* = P$ and satisfies $P^2 = P$. A projective measurement on $|\Psi\rangle$ consists of projectors $P_0, P_1, \dots, P_{M-1}$ such that $\sum_{k=0}^{M-1} P_k = I_N$ and $M \leq N$.

The outcome $m = k$ has probability $p_k = \langle \Psi|P_k|\Psi\rangle$

The post-measurement state is then $|\Psi_{\text{post}}^{(k)}\rangle = \frac{P_k|\Psi\rangle}{\sqrt{p_k}}$.

(This generalizes Von Neumann measurement, which is a special case where $N = M$).

In the 2-qubit case with $N = 4, M = 4$:

$$P_k = |k\rangle\langle k|, k = 0, 1, 2, 3$$

Then, we can verify that $P_k^* = P_k$, $P_k^2 = P_k$ and $\sum_{k=0}^3 P_k = I_4$, meaning that P_k is a projector.

Therefore $p_k = \langle \Psi|P_k|\Psi\rangle = \langle \Psi|k\rangle\langle k|\Psi\rangle = |a_k|^2$.

Then, the final state:

$$|\Psi_{\text{post}}^{(k)}\rangle = \frac{|k\rangle\langle k|\Psi\rangle}{|a_k|} = \frac{|k\rangle a_k}{|a_k|} = e^{j\theta a_k} |k\rangle \equiv |k\rangle \quad \text{complete measurement}$$

Note that the θ is the angle/phase of the qubit!

Partial Measurement

Ex: we have $|\Psi\rangle = |q_1\rangle \otimes |q_0\rangle$ (separable)

We want to measure $|q_0\rangle$ in computational basis!

We want to not change q_1 so we use a projector of the form

$$\begin{aligned} P(m=0) &= |\langle\Psi|I \otimes |0\rangle\langle 0|\Psi\rangle|^2 \\ &= |\langle q_1| \otimes \langle q_0|(I \otimes |0\rangle\langle 0|)|q_1\rangle \otimes |q_0\rangle|^2 \\ &= |\langle q_1|I|q_1\rangle|^2 |\langle q_0|0\rangle\langle 0|q_0\rangle|^2 \end{aligned}$$

Measuring $|q_0\rangle$ in $|q_1q_0\rangle$:

$$\begin{aligned} p(m=00) &= \langle\Psi|I \otimes |0\rangle\langle 0|\Psi\rangle|^2 \\ |\Psi_{\text{post}}^{(0)}\rangle &= \frac{I \otimes |0\rangle\langle 0|\Psi\rangle}{\sqrt{p(m=0)}} \end{aligned}$$

In the 2-qubit case measuring q_0 :

$N = 4, M = 2$ so:

- $P_0 = I \otimes |0\rangle\langle 0| = (|0\rangle\langle 0| + |1\rangle\langle 1|) \otimes |0\rangle\langle 0| = |00\rangle\langle 00| + |10\rangle\langle 10|$
- $P_1 = I \otimes |1\rangle\langle 1| = (|0\rangle\langle 0| + |1\rangle\langle 1|) \otimes |1\rangle\langle 1| = |01\rangle\langle 01| + |11\rangle\langle 11|$

Then:

$$\begin{aligned} P_0|\Psi\rangle &= a_{00}|00\rangle + a_{10}|10\rangle \implies p_0 = \langle\Psi|P_0|\Psi\rangle = |a_{00}|^2 + |a_{10}|^2 \\ \implies |\Psi_{\text{post}}^{(0)}\rangle &= \frac{a_{00}|00\rangle + a_{10}|10\rangle}{\sqrt{|a_{00}|^2 + |a_{10}|^2}} \end{aligned}$$

Similarly:

$$p_1 = |a_{01}|^2 + |a_{11}|^2 \implies |\Psi_{\text{post}}^{(1)}\rangle = \frac{a_{01}|01\rangle + a_{11}|11\rangle}{\sqrt{|a_{01}|^2 + |a_{11}|^2}}$$

Parity measurement

There are still two outcomes ($M = 2$).

- $\bar{P}_0 = |00\rangle\langle 00| + |11\rangle\langle 11|$
- $\bar{P}_1 = |01\rangle\langle 01| + |10\rangle\langle 10|$

To implement parity measurement, we have to change the basis:

- $|00\rangle \rightarrow |00\rangle$
- $|11\rangle \rightarrow |10\rangle$
- $|01\rangle \rightarrow |01\rangle$

- $|10\rangle \rightarrow |11\rangle$

This is simply rcX !

Key lesson: to perform different measurements, you have to change the basis/map the measurements to a basis that you CAN measure in! (remember to return the measurement to the original basis)

1.3 Entanglement

$|\Psi\rangle$ is entangled if it cannot be written as $|\Psi_1\rangle \otimes |\Psi_2\rangle$.

Bell states (aka ebits)

- $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$
- $|\Phi^-\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$
- $|\Psi^+\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$
- $|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$

e.g. $|\Phi^+\rangle$ is entangled:

$$\begin{aligned} |\Phi^+\rangle &= \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \\ &= (a_1|0\rangle + b_1|1\rangle) \otimes (a_0|0\rangle + b_0|1\rangle) \\ &= a_1a_0|00\rangle + a_1b_0|01\rangle + b_1a_0|10\rangle + b_1b_0|11\rangle \end{aligned}$$

This is a contradiction because of the definition of the states where only certain states have nonzero probability amplitudes.

Features:

- The bell states form an orthonormal basis for 2-qubit space - Bell basis.
- When one qubit is measured, the other qubit is completely determined:

If we measure $|q_0\rangle$ of $|\Phi^+\rangle$:

$$- P_0 = I \otimes |0\rangle\langle 0| \Rightarrow P_0|\Psi\rangle = \frac{1}{\sqrt{2}}|00\rangle \Rightarrow p_0 = \langle \Phi^+ | P_0 | \Phi^+ \rangle = \frac{1}{2}$$

$$|\Psi_{\text{post}}^{(0)}\rangle = \frac{(1/\sqrt{2})|00\rangle}{\sqrt{1/2}} = |00\rangle$$

$$- P_1 = I \otimes |1\rangle\langle 1|$$

- How do we generate Bell states?

Simply use an H gate on q_0 and CX with q_0 as control and q_1 as target.

$$\begin{aligned}
U = CX \cdot (I \otimes H) &= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & -1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 1 & 1 \\ 1 & -1 & 0 & 0 \end{bmatrix} \\
&= \frac{1}{\sqrt{2}} [|00\rangle + |11\rangle \quad |00\rangle - |11\rangle \quad |01\rangle + |10\rangle \quad -|01\rangle + |10\rangle] \\
&= [|\Phi^+\rangle \quad |\Phi^-\rangle \quad |\Psi^+\rangle \quad -|\Psi^-\rangle]
\end{aligned}$$

If we want to fix this, we can either use a SWAP gate after or we can use a CZ gate before (there may be other options).

We can also just look at the effect of the gates on the computational basis:

$$- |00\rangle \xrightarrow{I \otimes H} |0+\rangle \xrightarrow{CX} = |\Phi^+\rangle$$

- If we want to reverse the states, going from Bell to computational basis, we just apply CX and then $I \otimes H$.
- Transpose trick: why does it work? the math just works out (only for this Bell state)

$$(U \otimes I)|\Phi^+\rangle = (I \otimes U^T)|\Phi^+\rangle$$

LHS:

$$\frac{1}{\sqrt{2}} \left(\begin{bmatrix} u_{00} \\ u_{10} \end{bmatrix} \otimes |0\rangle + \begin{bmatrix} u_{10} \\ u_{11} \end{bmatrix} \otimes |1\rangle \right) = \frac{1}{\sqrt{2}} \begin{bmatrix} u_{00} \\ u_{01} \\ u_{10} \\ u_{11} \end{bmatrix}$$

RHS: do later

Superdense coding

Goal: transmit 2 classical bits by sending 1 qubit by using entanglement

Formulation:

- Alice and Bob each have a qubit in Bell state: ($q_1 = q_B$ and $q_0 = q_A$)

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

- Alice wants to send to Bob two classical bits $b_1 b_0 \in \{00, 01, 10, 11\}$
- Alice performs one of the 4 operations on her qubits depending on $b_1 b_0$

b_1b_0	U^A	$(I \otimes U^A) \Phi^+\rangle$
00	I	$ \Phi^+\rangle$
01	Z	$ \Phi^-\rangle$
10	X	$ \Psi^+\rangle$
11	ZX	$-\lvert\Psi^-\rangle$

- Alice then sends her qubit to Bob who then has both qubits.
- Finally, Bob measures the two qubits under Bell basis and uniquely determines b_1b_0 .

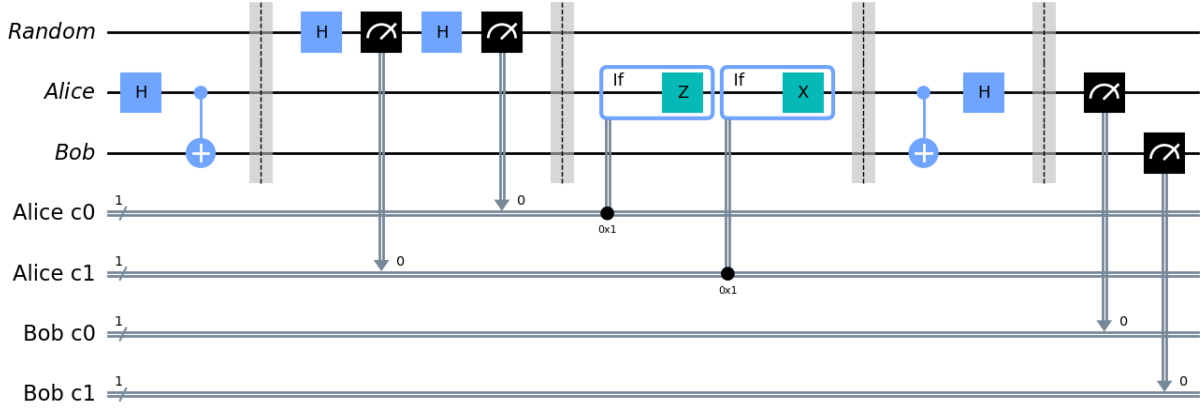


Figure 1.2: Superdense coding circuit

Teleportation

Sending two classical bits to send 1 qubit!

- Alice and Bob share an ebit (entangled qubit) $|\Phi^+\rangle_{BA}$.
- Alice wants to send a qubit $|\Psi\rangle_A = \alpha|0\rangle + \beta|1\rangle$.
- Alice combines $|\Psi\rangle_A$ with her share of $|\Phi^+\rangle_{BA}$.
- Alice measures her two qubits using Bell measurement and encodes the results using 2 classical bits.

$$\begin{aligned}
|\Phi^+\rangle_{BA} \otimes |\Psi\rangle_A &= \frac{1}{\sqrt{2}}[|00\rangle_{BA} + |11\rangle_{BA}] \otimes (\alpha|0\rangle_A + \beta|1\rangle_A) \\
&= \alpha|0\rangle_B \otimes \frac{1}{\sqrt{2}}|00\rangle_A \\
&\quad + \beta|1\rangle_B \otimes \frac{1}{\sqrt{2}}|11\rangle_A \\
&\quad + \beta|0\rangle_B \otimes \frac{1}{\sqrt{2}}|01\rangle_A \\
&\quad + \alpha|1\rangle_B \otimes \frac{1}{\sqrt{2}}|10\rangle_A \\
&= \frac{1}{2}(\alpha|0\rangle + \beta|1\rangle) \otimes |\Phi^+\rangle \quad \{|00\rangle = \frac{1}{2}[|\Phi^+\rangle + |\Phi^-\rangle]\} \\
&\quad + \frac{1}{2}(\alpha|0\rangle - \beta|1\rangle) \otimes |\Phi^-\rangle \quad \{|11\rangle = \frac{1}{2}[|\Phi^+\rangle - |\Phi^-\rangle]\} \\
&\quad + \frac{1}{2}(\beta|0\rangle + \alpha|1\rangle) \otimes |\Psi^+\rangle \quad \{|01\rangle = \frac{1}{2}[|\Psi^+\rangle + |\Psi^-\rangle]\} \\
&\quad + \frac{1}{2}(\beta|0\rangle - \alpha|1\rangle) \otimes |\Psi^-\rangle \quad \{|10\rangle = \frac{1}{2}[|\Psi^+\rangle - |\Psi^-\rangle]\} \\
&= \frac{1}{2}(|\Psi\rangle \otimes |\Phi^+\rangle + Z|\Psi\rangle \otimes |\Phi^-\rangle + X|\Psi\rangle \otimes |\Psi^+\rangle - ZX|\Psi\rangle \otimes |\Psi^-\rangle)
\end{aligned}$$

Therefore: To send some information, for Alice to communicate the state on the left, she sends the two classical bits, resulting in the following operations.

- $|\Phi^+\rangle \quad 00 \rightarrow I_B$
- $|\Phi^-\rangle \quad 01 \rightarrow Z_B$
- $|\Psi^+\rangle \quad 10 \rightarrow X_B$
- $|\Psi^-\rangle \quad 11 \rightarrow ZX_B$

Extension:

- Alice and charlie share a 2-qubit state $|\Psi\rangle_{CA}$.
- Alice teleports her share of $|\Psi\rangle_{CA}$ to Bob.
- Bob and Charlie share $|\Psi\rangle_{CB}$.

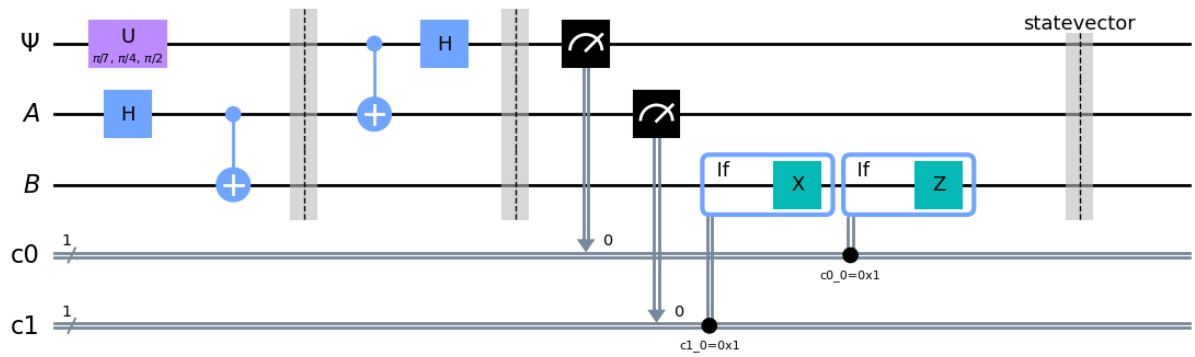


Figure 1.3: Caption

Chapter 2

Quantum Algorithms

Pure quantum algorithms - designing a unitary matrix and applying it to an initial state.

2.1 Quantum Oracle

2.1.1 Grover's search algorithm

Many real world problems are unstructured exhaustive searches!

Problem setup:

- Given a problem P and N data points $x_0, \dots, x_{N-1}$ an oracle is a binary function:

$$f(x_i) = \begin{cases} 1 & \text{if } x_i \text{ is a solution to } P, \\ 0 & \text{otherwise.} \end{cases}$$

- exhaustive unstructured search:
- classical approach has $O(N)$ complexity
- Quantum solver
 1. Map each datapoint x_k to a computational basis vector $|k\rangle_n$, therefore requiring $n = \lceil \log(N) \rceil$ qubits.
 2. Initialize the quantum circuit as $|h_0\rangle = H^{\otimes n}|0\rangle = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} |k\rangle_n$ (balanced superposition of all bases).
 3. Apply an oracle unitary:

$$U_f|k\rangle = \begin{cases} -|k\rangle, & \text{if } f(x_k) = 1 \\ |k\rangle, & \text{if } f(x_k) = 0. \end{cases} \Rightarrow U_f = \begin{bmatrix} (-1)^{f(x_0)} & & & \\ & (-1)^{f(x_1)} & & \\ & & \ddots & \\ & & & (-1)^{f(x_{N-1})} \end{bmatrix}_{N \times N}$$

4. Apply diffusion unitary operator to amplify the probability amplitude of the solution basis.
 5. Measure to get the solution to P .
- Typically, applying the diffusion operator once isn't enough, so you need to repeat steps 3 and 4 about $\sqrt{N}$ times before measuring at the end. This results in a complexity of $O(\sqrt{N})$, a quadratic speedup over the classical approach!

2.1.2 Oracle unitary operator

Instead of designing U_f , we design $\bar{U}_f$, with inputs $|x\rangle$ (n qubits) and $|y\rangle$ (1 ancilla bit). Output $|x\rangle$ (n qubits) and $|y \oplus f(x)\rangle$.

Checking that $\bar{U}_f$ is unitary:

$$\begin{aligned}\bar{U}_f[\bar{U}_f(|y\rangle \otimes |x\rangle)] &= \bar{U}_f[|y \oplus f(x)\rangle \otimes |x\rangle] = |y \oplus f(x) \oplus f(x)\rangle \otimes |x\rangle = |y\rangle \otimes |x\rangle \\ \implies \bar{U}_f^2 &= I.\end{aligned}$$

Now, we need to show that $\bar{U}_f^* = \bar{U}_f$ ($\bar{U}_f$ is real and $\bar{U}_f^T = \bar{U}_f$):

- By definition, each column of $\bar{U}_f$ is a computational basis vector of $(n+1)$ -qubits and it has a single 1 while the rest are 0s.
- To show that $\bar{U}_f$ is symmetric: we suppose that $m_{jk} = 0$ and $m_{kj} = 1$.

$$\begin{aligned}\bar{U}_f^2 = I &\Rightarrow \sum_l m_{jl} m_{lj} = 1, \quad (j, j)\text{-th element} \\ &= m_{jk} m_{kj} + \sum_{l \neq k} m_{jl} m_{lk} \\ &= 0 \times 1 + 0 \quad \text{by assumption} \\ &= 0 \quad \text{contradiction!}\end{aligned}$$

Input to U_f :

$$(H|1\rangle) \otimes (H^{\otimes n}|0\rangle) = |-\rangle \otimes \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} |k\rangle_n$$

Output to U_f :

$$\bar{U}_f \left[|-\rangle \otimes \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} |k\rangle_n \right] = \frac{1}{N} \sum_{k=0}^{N-1} \left(\frac{1}{\sqrt{2}} (|0 \oplus f(k)\rangle - |1 \oplus f(k)\rangle) \otimes |k\rangle \right)$$

- If $f(k) = 0$: $|y \oplus f(k)\rangle = |0\rangle - |1\rangle$
- If $f(k) = 1$: $|y \oplus f(k)\rangle = |1\rangle - |0\rangle$
- Therefore both are $(-1)^{f(x)}(|0\rangle - |1\rangle)$

This means the final output is:

$$\frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} (-1)^{f(k)} |k\rangle \otimes |-\rangle$$

This is a technique known as phase kickback!

- The phase was created in $|0\rangle$ but it ends up affecting the whole state!
- evaluate exponential number of function values with just one call of U_f - superposition!

Example: give tn the target basis, we can construct $\bar{U}_f$ by using just NOT and multicon-
trolled NOT gates.

2.1.3 Probability amplitude amplification (diffusion operator)

Example: $n = 2, N = 4$.

If we have the probability amplitudes: $a_0 = \frac{1}{2}, a_1 = \frac{1}{2}, a_2 = \frac{1}{2}, a_3 = \frac{1}{2}$, then $\bar{U}_f$ sets $a_1 = -\frac{1}{2}$.

We then apply the transformation $m - a_k$ where m is the mean.

$m = \frac{1}{N} \sum_{k=0}^{N-1} a_k = \frac{1}{4}$. This transforms $a_0 = a_2 = a_3 = 0$ and $a_1 = 1!!$ (This may often take more steps)

Inversion about the mean

Diffusion unitary

$$U_d = 2|h_o\rangle\langle h_o| - I_N, \quad \{|h\rangle = H^{\otimes n}|0\rangle, \text{ equal superposition}\}$$

This is a reflector matrix!

Need to show that this performs inversion about the mean!

$$\begin{aligned} U_d \left(\sum_{k=0}^{N-1} a_k |k\rangle \right) &= \sum_{k=0}^{N-1} a'_k |k\rangle, \quad \text{where } a'_k = 2m - a_k \\ U_d \left(\sum_{k=0}^{N-1} a_k |k\rangle \right) &= \sum_{k=0}^{N-1} a_k U_d |k\rangle = \sum_{k=0}^{N-1} a_k (2|h_o\rangle\langle h_o| - I) |k\rangle = 2\langle h_o|k\rangle |h_o\rangle - |k\rangle \\ &= \frac{1}{\sqrt{N}} \cdot \frac{1}{\sqrt{N}} \sum_{l=0}^{N-1} |l\rangle = \sum_k a_k \left[\left(\frac{2}{N} \sum_l |l\rangle \right) - |k\rangle \right] \\ &= \sum_l \left(\frac{2}{N} \sum_k a_k \right) |l\rangle - \sum_k a_k |k\rangle = \sum_k (2m - a_k) |k\rangle. \end{aligned}$$

You need to repeat the oracle and diffuser m times (will discuss what m is later)...

Circuit for Diffuser

$$\begin{aligned}
U_d &= 2|h_0\rangle\langle h_0| - I \\
\implies U_d|h_0\rangle &= 2|h_0\rangle\langle h_0|h_0\rangle - |h_0\rangle = |h_0\rangle. \\
\implies U_d|h_k\rangle &= -|h_k\rangle, \quad k \neq 0
\end{aligned}$$

Now in the computational basis:

$$\begin{cases} U_d H^{\otimes n} |0\rangle = H^{\otimes n} |0\rangle \\ U_d H^{\otimes n} |k\rangle = -H^{\otimes n} |k\rangle \end{cases} \implies \begin{cases} H^{\otimes n} U_d H^{\otimes n} |0\rangle = |0\rangle \\ H^{\otimes n} U_d H^{\otimes n} |k\rangle = -|k\rangle \end{cases} \quad \{(H^{\otimes n})(H^{\otimes n}) = I\}$$

For notational convenience let $H^{\otimes n} U_d H^{\otimes n} |k\rangle = U_0$.

$$U_0 = \begin{bmatrix} 1 & & & \\ & -1 & & \\ & & \ddots & \\ & & & -1 \end{bmatrix} = - \begin{bmatrix} -1 & & & \\ & 1 & & \\ & & \ddots & \\ & & & 1 \end{bmatrix} \quad \text{Letting } \bar{U}_0 = -U_0 \implies \begin{cases} \bar{U}_0 |0\rangle = -|0\rangle \\ \bar{U}_0 |k\rangle = -|k\rangle, k \neq 0 \end{cases}$$

We know that:

$$\text{MCZ} = \begin{bmatrix} 1 & & & \\ & \ddots & & \\ & & 1 & \\ & & & -1 \end{bmatrix} \implies \begin{cases} \text{MCZ}|k\rangle &= |k\rangle, k \neq N-1 \\ \text{MCZ}|N-1\rangle &= -|N-1\rangle \end{cases}$$

Note that $X^{\otimes n} |0\rangle = |N-1\rangle$. Therefore:

$$\begin{cases} \text{MCZ} X^{\otimes n} |0\rangle = -X^{\otimes n} |0\rangle \\ \text{MCZ} X^{\otimes n} |k\rangle = X^{\otimes n} |k\rangle \end{cases} \implies \begin{cases} X^{\otimes n} \text{MCZ} X^{\otimes n} |0\rangle = -|0\rangle \\ X^{\otimes n} \text{MCZ} X^{\otimes n} |k\rangle = |k\rangle \end{cases}$$

Then $\bar{U}_0 = -H^{\otimes n} U_d H^{\otimes n}$.

This gives us $U_d = -H^{\otimes n} X^{\otimes n} \text{MCZ} X^{\otimes n} H^{\otimes n}$

2.1.4 Geometric Interpretation

Suppose there are k targets (out of N). We define:

$$\begin{aligned}
|\alpha\rangle &= \frac{1}{\sqrt{N-k}} \sum_{f(x)=0} |x\rangle \\
|\beta\rangle &= \frac{1}{\sqrt{k}} \sum_{f(x)=1} |x\rangle
\end{aligned}$$

These are basis vectors and define the equal superpositions of the solution vectors and the nonsolution vectors. These are orthogonal, so we have an initial state:

$$|h_0\rangle = H^{\otimes n} |0\rangle = \sqrt{\frac{N-k}{N}} |\alpha\rangle + \sqrt{\frac{k}{N}} |\beta\rangle = \cos(\theta/2) |\alpha\rangle + \sin(\theta/2) |\beta\rangle$$

This vector $|h_0\rangle$ is in the space spanned by $|\alpha\rangle$ and $|\beta\rangle$.

When we apply the oracle:

$$\xrightarrow{\text{oracle}} U_f|h_0\rangle = \cos(\theta/2)|\alpha\rangle - \sin(\theta/2)|\beta\rangle.$$

This reflects $|h_0\rangle$ with respect to $|\alpha\rangle$. Then, when we apply U_d , we reflect across the original $|h_0\rangle$!!!

$$\xrightarrow{\text{diffuser}} U_d(U_f|h_0\rangle) = \text{reflect about } |h_0\rangle$$

This causes us to approach $|\beta\rangle$. After one iteration of $G = U_d U_f$:

$$G|h_0\rangle = \cos(3\theta/2)|\alpha\rangle + \sin(3\theta/2)|\beta\rangle$$

After m iterations the state becomes:

$$G^m|h_0\rangle = \cos\left(\frac{2m+1}{2}\theta\right)|\alpha\rangle + \sin\left(\frac{2m+1}{2}\theta\right)|\beta\rangle$$

Because we want the probability of amplitude of $|\beta\rangle$ to be 1, we choose m such that $(2m+1)\theta/2 \approx \pi/2$.

From the definition of the equal superposition, $\theta = 2\sin^{-1}\left(\sqrt{\frac{k}{N}}\right)$. Therefore, the final value of m is:

$$m = \frac{\pi}{2\theta} - \frac{1}{2}$$

For large values of n :

$$\begin{aligned} \Rightarrow \sin\sqrt{\frac{k}{N}} &\approx \sqrt{\frac{k}{N}} \\ \Rightarrow m &= \frac{\pi}{2\sqrt{\frac{k}{N}}} - \frac{1}{2} \\ &= \frac{\pi}{2}\sqrt{\frac{N}{k}} - \frac{1}{2} \end{aligned}$$

Therefore the complexity of Grover's is $O(\sqrt{N})$.

2.2 Deutsch-Jozsa Algorithm

2.2.1 Problem

Given a binary function $f : \{0,1\}^n \rightarrow \{0,1\}$, we know this function is either constant or balanced. Determine which type f is.

- constant: $f(x) = f(0)$ for all $x \in \{0, 1\}^n$
- balanced: half x takes $f(x) = 0$, the other half $f(x) = 1$.

Classical solution: evaluate half of x 's i.e. $2^{n-1} + 1$ queries.

Instead of exponential queries, we just use the oracle from Grover's algorithm to make 1 query, providing an exponential speedup!

2.2.2 DJ Circuit

Output of oracle

$$|\Psi_b\rangle = \frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} (-1)^{f(x)} |x\rangle$$

$$\xrightarrow{H^{\otimes n}} |\Psi_c\rangle = H^{\otimes n} |\Psi_b\rangle$$

Remember that:

$$H^{\otimes n} = [h_{l,k}] \rightarrow h_{l,k} = \frac{1}{\sqrt{N}} (-1)^{l \cdot k}$$

Therefore:

$$\begin{aligned} H^{\otimes n} |k\rangle &= \frac{1}{\sqrt{N}} \sum_{l=0}^{N-1} (-1)^{l \cdot k} |l\rangle \\ \Rightarrow H^{\otimes n} |\Psi_b\rangle &= \frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} (-1)^{f(x)} H^{\otimes n} |x\rangle \\ &= \frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} (-1)^{f(x)} \frac{1}{\sqrt{2^n}} \sum_{y \in \{0,1\}^n} (-1)^{x \cdot y} |y\rangle \\ &= \sum_{y \in \{0,1\}^n} \left(\frac{1}{2^n} \sum_{x \in \{0,1\}^n} (-1)^{f(x)} (-1)^{x \cdot y} \right) |y\rangle \end{aligned}$$

Consider the probability amplitude of $|0\rangle$ (i.e. $y = 0$):

$$a_0 = \frac{1}{2^n} \sum_{x \in \{0,1\}^n} (-1)^{f(x)} = \begin{cases} (-1)^{f(0)} & f(x) = f(0) \\ 0 & f(x) \text{ is balanced} \end{cases}$$

If the measurement has a zero (literally has any probability greater than 0), then it is constant, otherwise it is balanced.

2.2.3 Oracle unitaries

- constant:

$$\bar{U}_f(|x\rangle \otimes |y\rangle) = |x\rangle \otimes |y \oplus f(x)\rangle \Rightarrow |y \oplus f(x)\rangle = \begin{cases} |y\rangle & \text{if } f(x) = 0 \\ X|y\rangle & \text{if } f(x) = 1 \end{cases}$$

- balanced: example: $f(x) = x_{n-1} \oplus \bar{x}_{n-2} \oplus \dots \oplus x_0$

2.3 Bernstein-Vazirani Algorithm

2.3.1 The problem

Given a binary function $f(x) = a_{n-1}x_{n-1} + \dots + a_0x_0$ where $x_i, a_i \in \{0, 1\}$ find $\{a_i\}$ by querying $f(x)$.

Classical solution: Let $x_i = 1$ and $x_j = 0$ for all $j \neq i$ to get $a_i = f(x_i)$. This will require $O(n)$ queries.

Oracle output:

$$|\Psi_b\rangle = \frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} (-1)^{a \cdot x} |x\rangle = H^{\otimes n} |a\rangle \Rightarrow |\Psi_c\rangle = H^{\otimes n} |\Psi_b\rangle = |a\rangle$$

Speedup from $O(n) \rightarrow O(1)!$

2.4 Simon's Algorithm

The problem: A function $f : \{0, 1\}^n \rightarrow \{0, 1\}^n$ satisfies $f(x \oplus r) = f(x) \forall x \in \{0, 1\}^n$. The objective is to find r .

Assumptions:

- When $r = 0 \rightarrow f(x)$ is one-to-one.
- When $r \neq 0 \rightarrow f(x)$ is two-to-one.

Classical solution: there are 2^n possible r 's, the best algorithm has complexity $O(\sqrt{2^n})$.

We use the same grover-like circuit, but extend the oracle to use n ancilla bits for r .

2.4.1 Simon's circuit

n qubits and n ancilla (initialized as $|0\rangle$ instead of $|1\rangle$).

- First, after oracle, we measure the ancilla bits (to allow the input qubits to settle down).
- Then, we apply Hadamard gates to the input qubits

- Finally, we measure the input qubits for the final measurement.

States at each step:

1. Input to oracle:

$$|\Psi_a\rangle = H^{\otimes n}|0\rangle \otimes |0\rangle = \frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} |x\rangle$$

2. Output of oracle:

$$|\Psi_b\rangle = \frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} \bar{U}_f(|x\rangle \otimes |0\rangle)$$

- (a) Case 1: $r = 0 \rightarrow f(x)$ is one-to-one (letting $z = f(x)$ and $x = f^{-1}(z)$)

$$|\Psi_b\rangle = \frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} |x\rangle \otimes |f(x)\rangle = \frac{1}{\sqrt{2^n}} \sum_{z \in \{0,1\}^n} |f^{-1}(z)\rangle \otimes |z\rangle$$

Then, we perform partial measurement

$$P_z = I \otimes |z\rangle\langle z| \Rightarrow P_{z_0}|\Psi_b\rangle = \frac{1}{\sqrt{2^n}} |f^{-1}(z_0)\rangle \otimes |z_0\rangle$$

$$p(z_0) = \langle \Psi_b | P_{z_0} | \Psi_b \rangle = \frac{1}{2^n}, \quad z_0 \in \{0,1\}^n$$

$$|\Psi_c\rangle = \frac{P_{z_0}|\Psi_b\rangle}{\sqrt{p(z_0)}} = |f^{-1}(z_0)\rangle \otimes |z_0\rangle$$

Then, we apply Hadamard again (letting $f^{-1}(z_0) = x_0$ and $z_0 = f(x_0)$)

$$|\Psi_d\rangle = H^{\otimes n}|x_0\rangle = \frac{1}{\sqrt{2^n}} \sum_{y \in \{0,1\}^n} (-1)^{x_0 \cdot y} |y\rangle$$

If $r = 0$ then final measurement is uniform on $\{0,1\}^n$ ($p(y) = \frac{1}{2^n}$ with equal probability)!

- (b) Case 2: $r \neq 0 \rightarrow f(x)$ is two-to-one (range of $f : \{0,1\}^n \rightarrow \{0,1\}_f^n$, i.e. second set is a subset, has a size 2^{n-1}) and letting $z = f(x)$

$$|\Psi_b\rangle = \frac{1}{\sqrt{2^n}} \sum_{z \in \{0,1\}_f^n} (|x_z\rangle + |x_z + r\rangle) \otimes |z\rangle$$

Then, we perform partial measurement

$$P_{z_0}|\Psi_b\rangle = \frac{1}{\sqrt{2^n}} (|x_{z_0}\rangle + |x_{z_0} + r\rangle) \otimes |z_0\rangle$$

$$P_{z_0} = \langle \Psi_b | P_{z_0} | \Psi_b \rangle = \frac{1}{2^{n-1}}$$

$$|\Psi_c\rangle = \frac{P_{z_0}|\Psi_b\rangle}{\sqrt{P_{z_0}}} = \frac{1}{\sqrt{2}} (|x_{z_0}\rangle + |x_{z_0} + r\rangle) \otimes |z_0\rangle$$

Then we perform Hadamard gates on all input qubits

$$\begin{aligned}
|\Psi_d\rangle &= H^{\otimes n}|\Psi_c\rangle = \frac{1}{\sqrt{2}}(H^{\otimes n}|x_0\rangle + H^{\otimes n}|x_0 + r\rangle) \\
&= \frac{1}{\sqrt{2^{n+1}}} \sum_{y \in \{0,1\}} [(-1)^{x_0 \cdot y} + (-1)^{(x_0+r)y}] \cdot |y\rangle \\
&= \frac{1}{\sqrt{2^{n+1}}} \sum_{y \in \{0,1\}} [(-1)^{x_0 \cdot y} (1 + (-1)^{ry})] \cdot |y\rangle
\end{aligned}$$

In order to observe $|y\rangle$ (of course mod 2):

$$r \cdot y = 0.$$

Therefore, if we observe $y_1, \dots, y_m$ by solving $r \cdot y_i = 0$ for $i = 1, \dots, m$ we can get r ! If we run the circuit $n + k$ times, the probability of getting n linearly independent y_i 's is $1 - \frac{1}{2^{k+1}}$. Effectively, the complexity goes from $O^{n/2} \rightarrow O(n)$.

Alternatively, we can directly apply $H^{\otimes n}$ at the output of the oracle $\bar{U}_f$ and then measure.

$$\rightarrow P_y = |y\rangle\langle y| \otimes I$$

The probability of observing a particular $|y\rangle$ is:

- if $r = 0 \Rightarrow p_y = \frac{1}{2^n}$
- if $r \neq 0 \Rightarrow p_y = 0$ if $ry = 1$ and $p_y = \frac{1}{2^{n-1}} = 0$ if $ry = 0$.

2.4.2 Oracle

Construct $f(x)$ such that $f(x) = f(x \oplus r), r \neq 0$.

- If $r \neq 0$, let j be the least index such that $r_j = 1$.
- For any $x \in \{0,1\}^n$ define

$$g(x) = \begin{cases} x & \text{if } x_j = 0 \\ x \oplus r & \text{if } x_j = 1 \end{cases}$$

then $g(x) = g(x \oplus r)$.

Proof. By definition

$$g(x \oplus r) = \begin{cases} x \oplus r & \text{if } x_j + r_j = 0 \leftrightarrow x_j = 1 \\ x \oplus r \oplus r = x & \text{if } x_j + r_j = 1 \leftrightarrow x_j = 0 \end{cases} = g(x)$$

Thus $g(x)$ is a period function (then we can permute $g(x)$ to get $f(x)$). □

Example: $r = 101$

2.5 Quantum Fourier Transform

2.5.1 QFT

1. Classical DFT and IDFT:

Given

$$|\Psi\rangle = \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_{N-1} \end{bmatrix} \in \mathbb{C}^N = \sum_{k=0}^{N-1} a_k |k\rangle,$$

the DFT of Ψ is

$$\text{DFT}\{|\Psi\rangle\} = |\tilde{\Psi}\rangle = \begin{bmatrix} \tilde{a}_0 \\ \tilde{a}_1 \\ \vdots \\ \tilde{a}_{N-1} \end{bmatrix} \in \mathbb{C}^N = \sum_{k=0}^{N-1} \tilde{a}_k |k\rangle, \quad \tilde{a}_k = \frac{1}{\sqrt{N}} \sum_{l=0}^{N-1} e^{j\frac{2\pi}{N}kl} a_l, \quad k = 0, \dots, N-1$$

DFT matrix:

$$W_N = \frac{1}{\sqrt{N}} \begin{bmatrix} e^{j\frac{2\pi}{N}kl} a_l \end{bmatrix} = \frac{1}{\sqrt{N}} \begin{bmatrix} 1 & 1 & \dots & 1 \\ 1 & \omega & \dots & \omega^{N-1} \\ 1 & \omega^2 & \dots & \omega^{2(N-1)} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & \omega^{N-1} & \dots & \omega^{(N-1)(N-1)} \end{bmatrix}_{N \times N}, \quad \omega = e^{j\frac{2\pi}{N}}$$

FACT: W_N is unitary!

DFT: $|\tilde{\Psi}\rangle = W_N \cdot |\Psi\rangle$ has complexity $O(N^2)$ but FFT allows for $O(N \cdot \log N)$

IDFT: $|\Psi\rangle = W_N^* |\tilde{\Psi}\rangle$

$$\begin{aligned} |\tilde{\Psi}\rangle &= W_N |\Psi\rangle = \sum_{k=0}^{N-1} \tilde{a}_k |k\rangle - \text{classical} \\ &= \sum_{k=0}^{N-1} a_k \cdot (W_N |k\rangle) \\ &= \sum_{k=0}^{N-1} a_k |\tilde{k}\rangle - \text{quantum} \end{aligned}$$

In quantum, we vary the basis! To do this, we find W_N :

$$W_N = [|0\rangle, |1\rangle, \dots, |N-1\rangle] = [|\tilde{0}\rangle, |\tilde{1}\rangle, \dots, |\tilde{N-1}\rangle], \quad \{\text{Fourier basis}\}$$

2. Fourier basis:

$$\begin{aligned}
x &= (x_{n-1}, x_{n-2}, \dots, x_0)_2 \\
&= x_{n-1} \cdot 2^{n-1} + \dots + x_1 \cdot 2 + x_0 \\
y &= (y_{n-1}, y_{n-2}, \dots, y_0)_2 \\
&= y_{n-1} \cdot 2^{n-1} + \dots + y_1 \cdot 2 + y_0
\end{aligned}$$

Then:

$$\begin{aligned}
|\tilde{x}\rangle &= \frac{1}{\sqrt{N}} \sum_{y=0}^{N-1} e^{j\frac{2\pi}{N}xy} |y\rangle \\
&= \frac{1}{\sqrt{N}} \sum_{y=0}^{N-1} e^{j\frac{2\pi}{N}x \cdot \frac{y_{n-1}2^{n-1} + \dots + y_0}{2^n}} |y_{n-1} \dots y_0\rangle \\
&= \frac{1}{\sqrt{N}} \sum_{y=0}^{N-1} e^{j2\pi x \cdot \sum_{k=0}^{N-1} \frac{y_k}{2^{n-k}}} |y_{n-1} \dots y_0\rangle \\
&= \sum_{y \in \{0,1\}^N} \prod_{k=0}^{N-1} \frac{1}{\sqrt{2}} e^{j2\pi \frac{xy_k}{2^{n-k}}} |y_{n-1} \dots y_0\rangle \\
&= \sum_{y \in \{0,1\}^N} \bigotimes_{k=0}^{N-1} \frac{1}{\sqrt{2}} e^{j2\pi \frac{xy_k}{2^{n-k}}} |y_k\rangle \\
&= \sum_{y_{n-1}, y_{n-2}, \dots, y_1} \bigotimes_{k=1}^{N-1} \frac{1}{\sqrt{2}} e^{j2\pi \frac{xy_k}{2^{n-k}}} |y_k\rangle \otimes \frac{1}{\sqrt{2}} (|0\rangle + e^{j2\pi \frac{x}{2^n}} |1\rangle) \quad \text{unrolling } y_0 \in \{0,1\} \\
&= \sum_{y_{n-1}, y_{n-2}, \dots, y_2} \bigotimes_{k=2}^{N-1} \frac{1}{\sqrt{2}} e^{j2\pi \frac{xy_k}{2^{n-k}}} |y_k\rangle \otimes \frac{1}{\sqrt{2}} (|0\rangle + e^{j2\pi \frac{x}{2^n}} |1\rangle) \otimes \frac{1}{\sqrt{2}} (|0\rangle + e^{j2\pi \frac{x}{2^{n-1}}} |1\rangle) \\
&= \text{continue doing this ...}
\end{aligned}$$

Recall that the P -gate:

$$P(\theta) = \begin{bmatrix} 1 & 0 \\ 0 & e^{j\theta} \end{bmatrix}$$

Such that

$$\begin{cases} P(\theta)|0\rangle = |0\rangle \\ P(\theta)|1\rangle = e^{j\theta}|1\rangle \end{cases} \Rightarrow P(\theta)|+\rangle = P(\theta) \cdot \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = \frac{1}{\sqrt{2}}(|0\rangle + e^{j\theta}|1\rangle)$$

Therefore:

$$\begin{aligned}
|\tilde{x}\rangle &= |x_{n-1}\rangle \otimes \dots \otimes |\tilde{x}_1\rangle \otimes |\tilde{x}_0\rangle \\
&= P(2\pi \frac{x}{2})|+\rangle \otimes \dots \otimes P(2\pi \frac{x}{2^{n-1}})|+\rangle \otimes P(2\pi \frac{x}{2^n})|+\rangle
\end{aligned}$$

3. Bloch sphere interpretation of QFT

$$|\Psi\rangle = \cos(\theta/2)|0\rangle + \sin(\theta/2)e^{j\phi}|1\rangle$$

(We only perform QFT on the computational basis, so we have $\theta/2$ to construct the plus state)

Fourier transform maps each qubit $|0\rangle$ or $|1\rangle$ on the Z axis to the XY plane with a rotation angle of $\phi_i = \frac{k}{2^{n-i}}2\pi$.

In the case $n = 3$ from k to $k + 1$:

- $\phi_0 = \frac{2\pi}{8}$
- $\phi_1 = \frac{2\pi}{4}$
- $\phi_2 = \frac{2\pi}{2}$

Example: $n = 3$ and $k = 5 = (101)_2$

- $\phi_0 = 2\pi \frac{5}{8} = \frac{5\pi}{4}$
- $\phi_1 = 2\pi \frac{5}{4} = \frac{5\pi}{2} \equiv \pi/2 \pmod{2\pi}$
- $\phi_2 = 2\pi \frac{5}{2} = 5\pi \equiv \pi \pmod{2\pi}$

4. QFT circuit:

Why does this equal the QFT?

$$H|x\rangle = \frac{1}{\sqrt{2}}(|0\rangle + (-1)^x|1\rangle), \quad (-1) = e^{j\pi}$$

See in the circuit: When we apply H to $|x_{n-1}\rangle$:

$$\begin{aligned} l = n - 1 &\xrightarrow{H} \frac{1}{\sqrt{2}}(|0\rangle + e^{j2\pi \frac{x_{n-1}}{2}}|1\rangle) \\ k = 1 &\xrightarrow{CP_{n-2}(\pi/2)} \frac{1}{\sqrt{2}}(|0\rangle + e^{j2\pi(\frac{x_{n-1}}{2} + \frac{x_{n-2}}{2^2})}|1\rangle) \\ k = n - 1 &\xrightarrow{CP_0(\pi/2)} \frac{1}{\sqrt{2}}(|0\rangle + e^{j2\pi(\frac{x_{n-1}}{2} + \dots + \frac{x_0}{2^n})}|1\rangle) = \frac{1}{\sqrt{2}}(|0\rangle + e^{j2\pi \frac{1}{2^n}(x_{n-1}2^{n-1} + \dots + x_0)}|1\rangle) \\ &= \frac{1}{\sqrt{2}}(|0\rangle + e^{j2\pi \frac{x}{2^n}}|1\rangle) \end{aligned}$$

Similarly:

- $|x_{n-1}\rangle \rightarrow P(2\pi x/2^n)|+\rangle$
- $|x_{n-2}\rangle \rightarrow P(2\pi x/2^{n-1})|+\rangle$
- $|x_0\rangle \rightarrow P(2\pi x/2)|+\rangle$

This gives the QFT $\{x\}$ with the order of qubits reversed, so you SWAP them all!

Complexity of QFT: $(n-1) + (n-2) + \dots + 1 = O(n^2) = O((\log N)^2)$

The inverse QFT you just reverse the QFT circuit!

Note, the output of this circuit is separable (as well as the input).

2.5.2 Quantum Phase Estimation

The problem

Given a unitary U and its eigenvector $|\Psi\rangle$, then we can write

$$U|\Psi\rangle = \lambda|\Psi\rangle = e^{j2\pi\theta}|\Psi\rangle, \theta \in [0, 1]$$

We can write $\lambda = e^{j2\pi\theta}$ because U is unitary.

Suppose that $2^n\theta = x = (x_{n-1}, \dots, x_1, x_0)_2 \Rightarrow e^{j2\pi\theta} = e^{j\frac{2}{2^n}x}$

QPE circuit

n qubits initialized to 0 and then we apply H . The last qubit is initialized to eigenvector $|\Psi\rangle$. We then apply the U gate, controlled by each qubit, onto $|\Psi\rangle$. The first qubit controls $U^{2^{n-1}}$, the second controls $U^{2^{n-2}}$, until U^{2^0} . Then, we apply the inverse QFT and measure all of the states. This has depth $2^{n-1} + 2^{n-2} + \dots + 2^0 = 2^n - 1$.

What happens if we apply CU^{2^k} ?

$$\begin{aligned} & CU^{2^k}(|+\rangle \otimes |\Psi\rangle) \\ &= \frac{1}{\sqrt{2}}[|0\rangle \otimes |\Psi\rangle + |1\rangle \otimes U^{2^k}|\Psi\rangle] \\ &= \frac{1}{\sqrt{2}}[|0\rangle \otimes |\Psi\rangle + |1\rangle \otimes e^{j2\pi(2^k)\theta}|\Psi\rangle] \\ &= \frac{1}{\sqrt{2}}[|0\rangle + e^{j2\pi\frac{x}{2^{n-k}}} |1\rangle] \otimes |\Psi\rangle \\ &= P\left(\frac{x}{2^{n-k}}\right) |+\rangle \otimes |\Psi\rangle \end{aligned}$$

Looking back at the circuit, after each control:

$$\begin{aligned}
& \xrightarrow[k=n-1]{U^{2^{n-1}}} |\Psi_1\rangle = P(2\pi \frac{x}{2})|+\rangle \otimes |+\rangle^{\otimes(n-1)} \otimes |\Psi\rangle \\
& \xrightarrow[k=n-2]{U^{2^{n-2}}} |\Psi_2\rangle = P(2\pi \frac{x}{2})|+\rangle \otimes P(2\pi \frac{x}{2^2})|+\rangle \otimes |+\rangle^{\otimes(n-2)} \otimes |\Psi\rangle \\
& \quad \dots \\
& \xrightarrow[k=0]{U^{2^0}} |\Psi_n\rangle = P(2\pi \frac{x}{2})|+\rangle \otimes P(2\pi \frac{x}{2^2})|+\rangle \otimes \dots \otimes P(2\pi \frac{x}{2^n})|+\rangle \otimes |\Psi\rangle
\end{aligned}$$

This is just QFT $\{x\}$!

Example: P-gate: $P(\theta)|1\rangle = e^{j\theta}|1\rangle$

Therefore, this circuit allows us to estimate θ .

Example: Pauli rotation gates

$$\begin{aligned}
R_q(\theta) &= \exp\left(-j\frac{\theta}{2}q\right), q \in \{X, Y, Z\} \\
q &= |v_0\rangle\langle v_0| - |v_1\rangle\langle v_1|
\end{aligned}$$

Applying either of the eigenvectors:

$$\begin{aligned}
R_q(\theta)|v_0\rangle &= e^{-j\frac{\theta}{2}}|v_0\rangle \\
R_q(\theta)|v_1\rangle &= e^{j\frac{\theta}{2}}|v_1\rangle
\end{aligned}$$

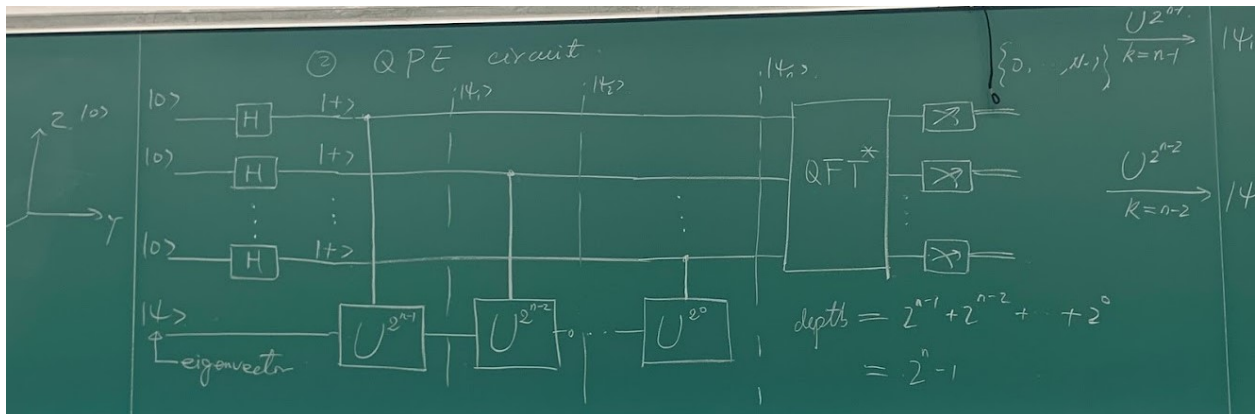


Figure 2.1: QPE

2.5.3 Basic Arithmetic with QFT

First step: phase encoding of integer l (using QFT) using m qubits

$$\begin{aligned} |\tilde{l}\rangle_m &= QFT\{|l\rangle_m\} = \frac{1}{\sqrt{2^m}} e^{j\frac{2\pi}{2^m}lk} |k\rangle_m \quad \{\text{by definition}\} \\ |l + \tilde{2^m}\rangle &= |\tilde{l}\rangle_m \quad \text{can only represent } 2^m \text{ numbers} \end{aligned}$$

The special case $|\tilde{0}\rangle = \frac{1}{\sqrt{2^m}} \sum |k\rangle_m = H^{\otimes m} |0\rangle_m$. We need a unitary $U(p)$ such that $U(p)|\tilde{l}\rangle_m = |l + \tilde{p}\rangle_m$.

To do this, we start with zeros in the computational basis, convert to QFT basis using hadamard gates, apply $U(l)$ and $U(p)$ to get $|l + \tilde{p}\rangle_m$ and then use the inverse QFT to measure. For this to work, we need for the numbers to be within the range:

$$-2^{m-1} \leq l, p, l + p \leq 2^{m-1} - 1$$

Two's complement: represent any integer $x \in [-2^{m-1}, 2^{m-1} - 1]$ using m bits.

- if $x > 0$, use the binary expansion of x
- if $x < 0$, use the binary expansion of $2^m + x$

The advantage of this is that regular addition becomes regular binary addition by discarding the last carry-out bit (if there is any)

We just apply a rotation to each qubit $P(\frac{2\pi}{2^{m-i}}p)$!

Proposition 1:

$$U(p)|k\rangle = e^{j\frac{2\pi}{2^m}pk} |k\rangle$$

Consider the value of q_i :

$$|q_i\rangle \rightarrow P(\frac{2\pi}{2^{m-i}}p)|q_i\rangle = \begin{cases} |0\rangle, q_i = 0 \\ e^{j\frac{2\pi}{2^m}2^i p} |1\rangle, q_i = 1 \end{cases} = e^{j(\frac{2\pi}{2^m}2^i q_i p)} |q_i\rangle$$

This gives us:

$$\begin{aligned} U(p)|k\rangle &= \bigotimes_{i=0}^{m-1} P(\frac{2\pi}{2^{m-i}}p)|q_i\rangle \\ &= \bigotimes_{i=0}^{m-1} e^{j(\frac{2\pi}{2^m}2^i q_i p)} |q_i\rangle \\ &= e^{j(\frac{2\pi}{2^m} \sum_{i=0}^{m-1} q_i p)} \bigotimes_{i=1}^{m-1} |q_i\rangle \\ &= e^{j\frac{2\pi}{2^m}pk} |k\rangle \end{aligned}$$

Proposition 2:

$$U(p)|\tilde{l}\rangle = |l + \tilde{p}\rangle$$

$$U(p)|\tilde{l}\rangle = \frac{1}{\sqrt{2^m}} \sum_{k=0}^{2^m-1} e^{j(\frac{2\pi}{2^m}lk)} U(p)|k\rangle = \frac{1}{\sqrt{2^m}} \sum_{k=0}^{2^m-1} e^{j(\frac{2\pi}{2^m}(l+p)k)} |k\rangle = |l+p\rangle_m$$

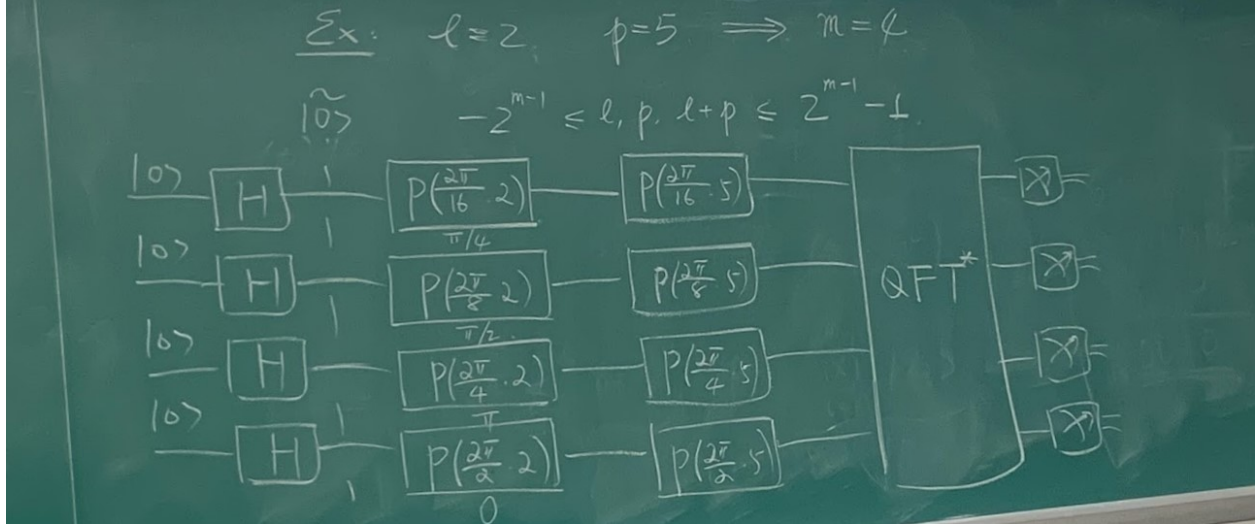


Figure 2.2: Arithmetic in QFT basis

2.5.4 Computing binary polynomials

Example: Binary quadratic function

$$f(x) = \sum_{i,j} a_{ij}x_i x_j + \sum_i b_i x_i + c, \quad x_i \in \{0,1\}$$

- n : number of binary variables x_i
 - m : number of qubits for computing $f(x)$
1. We need to choose m such that $(-2^{m-1} \leq k^-, k^+ \leq 2^{m-1} - 1$ where k^+ is the sum of positive coefficients and k^- is the sum of negative coefficients.
 2. binary multiplication: controlled addition

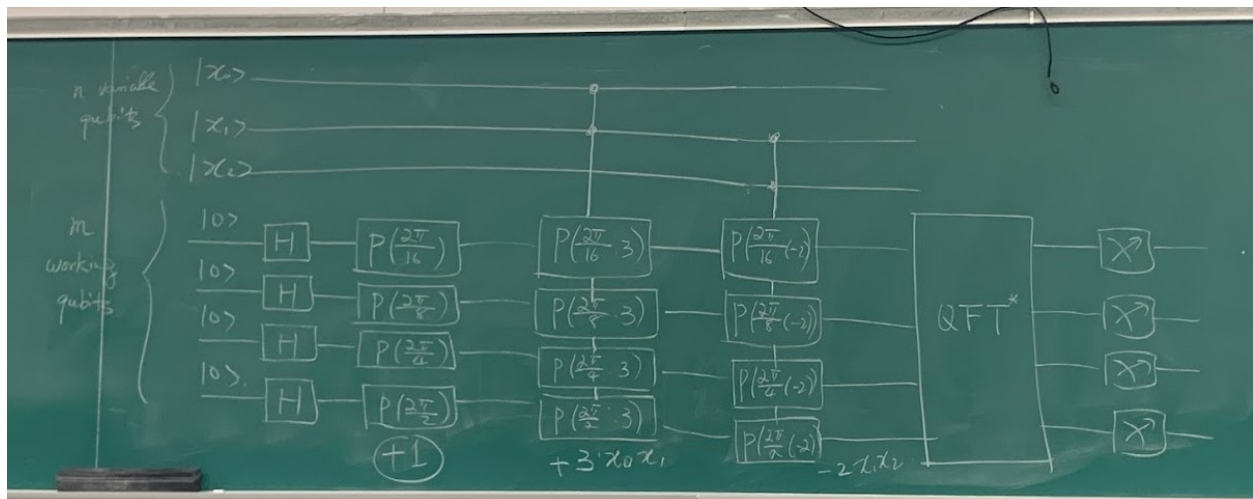


Figure 2.3: Computing polynomial in QFT basis

Chapter 3

Quantum Optimization

3.1 QUBO and GAS

3.1.1 Quadratic Unconstrained Binary Optimization

Max-cut

Given a graph, $G = (V, E)$, divide the vertex set V into $V = V_1 \cup V_2$, $V_1 \cap V_2 = \emptyset$ such that the size of the cut set $C = \{(v_1, v_2) \in E, v_1 \in V_1, v_2 \in V_2\}$ is maximized.

let $z_i \in \{-1, 1\}$ such that

$$z_i = \begin{cases} -1 \Rightarrow v_i \in V_1 \\ 1 \Rightarrow v_i \in V_2 \end{cases}$$

the $(v_i, v_j) \in E$ is a cut iff $z_i z_j = -1$.

max-cut prob:

$$\min \sum_{(i,j) \in E} z_i z_j, z_i \in \{-1, 1\}$$

notes:

- binary
- unconstrained

0-1 linear program

$$\begin{aligned} \min & -5x_0 + 3x_1 - 2x_2 \\ \text{s.t.} & x_0 + x_2 \leq 1 \\ & 3x_0 - x_1 + 3x_2 \leq 4 \\ & x_i \in \{0, 1\} \end{aligned}$$

To remove the inequality constraints we can introduce slack variables and add penalty terms.

- For slack variables, note that each variable can only take on the values of 0 or 1, so to have a slack variable that goes between 0 and 5, you might need to add a variable $y_1 + 2y_2 + 3y_3$

Then, we incorporate the constraints as a penalty in terms in the objective.

$$\min -5x_0 + 3x_1 - 2x_2 + B(x_0 + x_2 + y_0 - 1)^2 + B(3x_0 - x_1 + 3x_2 + y_1 + 2y_2 + 2y_3 - 4)^2$$

We choose B large enough to force the constraints to go to 0. The range of objectives is $[-7, 3]$, so if $-7 + B \geq 3$, we are no longer in a feasible result.

Knapsack problem

m objects $j = 1, \dots, m$ and each with weight w_j and value c_j . We define

$$x_i = \begin{cases} 1 & \text{if objective } i \text{ is selected} \\ 0 & \text{otherwise.} \end{cases} \Rightarrow \max \sum_{i=1}^m c_i x_i - \text{totalvalue}$$

subject to $\sum_{i=1}^m w_i x_i \leq W$

Travelling salesman problem

m cities, distance between i and j , w_{ij} .

Find path that visits each city only once that minimizes the total distance. We can define

$$x_{il} = \begin{cases} 1 & \text{if city } i \text{ is the } l\text{-th stop} \\ 0 & \text{otherwise.} \end{cases}$$

Constraints:

- Each city is visited once: $\sum_{l=1}^m x_{il} = 1$ for each i .
- At each stop, only one city is visited $\sum_{i=1}^m x_{il} = 1$ for each l .

The cost of $i \rightarrow j$: $i \rightarrow j$ occurs iff $x_{il}x_{j,l+1} = 1$ for some l . Otherwise, $x_{il}x_{j,l+1} = 0$.

Therefore, the total cost is:

$$\min \sum_l \sum_i \sum_j w_{ij} x_{il} x_{j,l+1}$$

Solving these problems

Grover's search: (grover adaptive search)

- initial $x_0 \rightarrow$ find all x such that $g(x) \leq g(x_0) \rightarrow x_1$
- iterate with x_1 and onwards...

Binary Optimization (quadratic)

$$\begin{aligned} & \min g(x) \\ & \text{max-cut : } x \in \{+1, -1\}^n \quad \text{Hamiltonian} \\ & \text{knapsack, TSP : } x \in \{0, 1\}^n - \text{QUBO, HOBQ} \end{aligned}$$

(HOBQ refers to high-order)

We can map these two forms:

$$\begin{aligned} x \in \{0, 1\} & \leftrightarrow z \in \{+1, -1\} \\ z &= 1 - 2x \\ x &= \frac{1 - z}{2} \end{aligned}$$

3.1.2 Grover Adaptive Search (GAS)

Consider quadratic unconstrained binary optimization or high-order binary optimization problem:

$$\min g(x), \quad x \in \{0, 1\}^n$$

Oracle

At the i -th iteration, for a given x_i :

$$f^{(i)}(x) = \begin{cases} 0 & \text{if } g(x) \geq g(x_i) \\ 1 & \text{if } g(x) < g(x_i) \end{cases}$$

Also, let r_i represent the number of Grover iterations at outer iteration i .

- choose the number of grover iterations r_i at random over a range that is increasing. $\Rightarrow$ can find a target w/ prob. within $O(\sqrt{2^n})$ iterations.

Recall Grover Circuit...

Oracle for $f^{(i)}(x)$ For input x of size n and m qubits to perform addition on, we need to first compute $g(x) - y_i$ (think hw 5 problem 3) and apply inverse QFT to bottom bits. If the most significant bit is 1 (i.e. $g(x) - y_i < 0$) then we flip the ancillary bit to satisfy $y \oplus f(x)$. Then we perform uncomputation to decontaminate counting qubits by performing reverse of gates.

The idea

Classical vs Quantum

$$\begin{aligned} \min g(x) & \leftrightarrow \min g(|x\rangle) \\ x \in \{0, 1\}^n \subset \mathbb{C}^n & \leftrightarrow |x\rangle = \{|0\rangle_n, \dots, |N-1\rangle_n\} \subset \mathbb{C}^{2^n} \end{aligned}$$

Out of $N = 2^n$ possible, superposition allows us to achieve (effectively) parallelism to find our measurement more quickly.

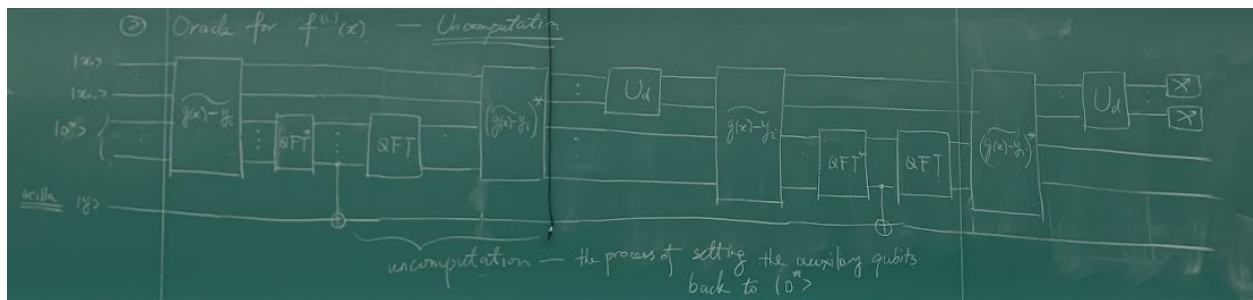


Figure 3.1: GAS Oracle

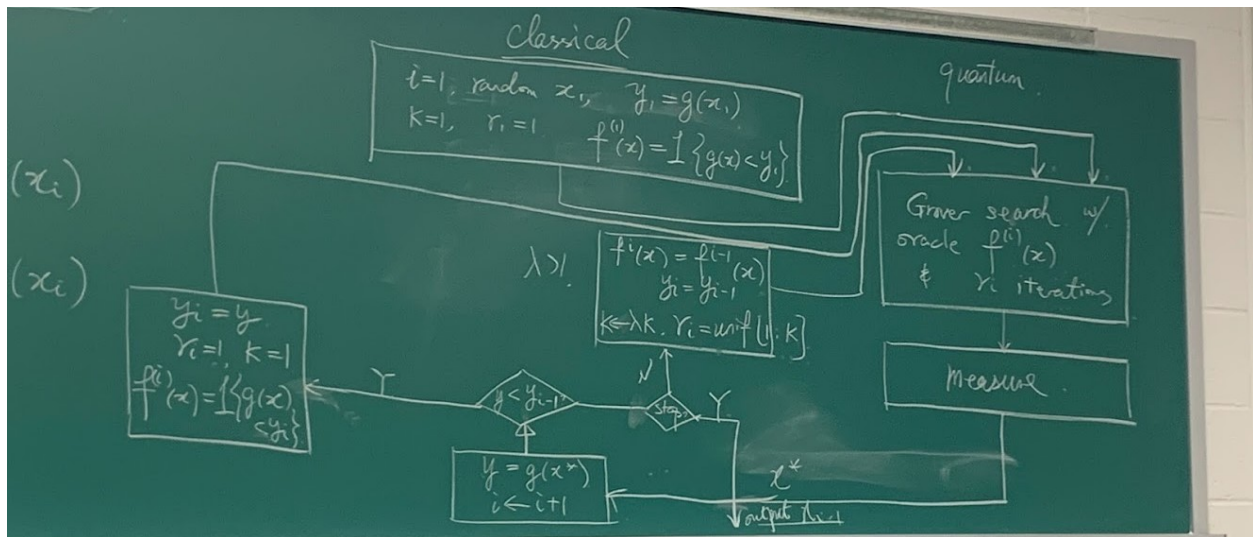


Figure 3.2: GAS Algorithm

3.2 Quantum Annealing

Ising Model

- For modeling interactions of particle spins.
- binary variables z_i for particle i :

$$z_i = \begin{cases} +1 & \text{spin up} \\ -1 & \text{spin down} \end{cases}$$

Can map $+1$ to $|0\rangle$ and -1 to $|1\rangle$.

- total energy - Hamiltonian

$$H(z) = - \sum_{i \neq j} J_{i,j} z_i z_j - \sum_i h_i z_i$$

J_{ij}, h_i : real coefficients. $\implies$ we want to find the state of min energy $\min H(z), z \in \{+1, -1\}$.

Special case: $J_{ij} = -1, h_i = 0$ (min energy is the same as min-cut) and z_i is QUBO $x \in \{0, 1\}$?

or alternatively:

$$H = \sum_{i \neq j} a_{ij} Z_{ij} + b_i Z_i$$

where the Pauli strings are:

$$\begin{aligned} Z_{ij} &= I \otimes \dots \otimes Z \otimes \dots \otimes Z \otimes \dots \otimes I \\ Z_i &= I \otimes \dots \otimes Z \otimes \dots \otimes I \end{aligned}$$

Quantum Formulation

Ex: max-cut of wedge with three nodes, 0 is connected with 2 and 1.

$$\min z_0 z_1 + z_0 z_2, \quad z_i \in \{+1, -1\}$$

1. Map z_i to qubit $|x_i\rangle$ such that

$$\begin{aligned} z_i = 1 &\rightarrow |x_i\rangle = |0\rangle \\ z_i = -1 &\rightarrow |x_i\rangle = |1\rangle. \end{aligned}$$

2. Recall the Z operator

$$\begin{aligned} Z &= |0\rangle\langle 0| - |1\rangle\langle 1| \\ \implies \begin{cases} \langle 0|Z|0\rangle = 1 \\ \langle 1|Z|1\rangle = -1 \end{cases} \end{aligned}$$

Let us define $Z_{01} = I \otimes Z \otimes Z$.

$$\begin{aligned} &\implies \langle x|Z_{01}|x\rangle \\ &= \langle x_2|I|x_2\rangle \cdot \langle x_1|Z|x_1\rangle \cdot \langle x_0|Z|x_0\rangle \\ &= z_0 z_1 \end{aligned}$$

Similarly,

$$Z_{ij} = z_i z_j$$

Therefore, our problem is translated to the quantum circuit where $|x\rangle$ is a computational basis.

$$\min \langle x|Z_{01} + Z_{02}|x\rangle$$

Recall the Ising model: we define:

$$|x\rangle = |x_{n-1}\rangle \otimes \dots \otimes |x_0\rangle \quad |x_i\rangle = \left| \frac{1 - z_i}{2} \right\rangle$$

$$Z_i = \bigotimes_{l=0}^{n-1} U_l \quad \text{with } U_l = \begin{cases} I & \text{if } l \neq i \\ Z & \text{if } l = i \end{cases}$$

$$Z_{ij} = \bigotimes_{l=0}^{n-1} V_l \quad \text{with } V_l = \begin{cases} I & \text{if } l \neq i, l \neq j \\ Z & \text{if } l = i \text{ or } l = j \end{cases}$$

This allows us to transform the Ising model into quantum form (this operator is Hermitian aka observable)

$$H = - \sum_{i \neq j} J_{ij} Z_{ij} - \sum_i h_i Z_i$$

Therefore, to minimize the energy in our system:

$$\min \langle x | H | x \rangle = \langle H \rangle_{|x\rangle}$$

This gives us the ground state $|\Psi^*\rangle$ which is the minimization of the problem.

Observability

Recall:

- Von Neumann measurement

With orthonormal basis: $|V_x\rangle, n = 0, \dots, N-1$. We can measure outcome x w.p. $|\langle v_x | \Psi \rangle|^2$.

- Projective measurement

Let $X_y, y = 0, 1, \dots, N'-1$ where $N' < N$ be a partition of $\{0, \dots, N-1\}$. The y -th projector is:

$$\Pi_y = \sum_{x \in X_y} |V_x\rangle \langle V_x|$$

This gives us the outcome y with probability $\langle \Psi | \Pi_y | \Psi \rangle$, where $y = 0, \dots, N'-1$.

- Observable: assign a real value $O_y \in \mathbb{R}$ to each projector $\Pi_y, y = 0, \dots, N'-1$.

Then, the outcome O_y occurs with probability $\langle \Psi | \Pi_y | \Psi \rangle$. This is a random variable!

$$\mathbb{E}\{\text{outcome}\} = \sum_{y=0}^{N'-1} O_y \langle \Psi | \Pi_y | \Psi \rangle = \left\langle \Psi \left| \sum_{y=0}^{N'-1} O_y \Pi_y \right| \Psi \right\rangle = \langle O \rangle_{|\Psi\rangle}$$

We define $\sum_{y=0}^{N'-1} O_y \Pi_y$ as O - observable and hermitian. Then, this summation provides the eigendecomposition of the matrix. Any expected value of observable O w.r.t state $|\Psi\rangle$.

Therefore, any Hermitian matrix O is an observable because

$$O = \sum_{x=0}^{N-1} \lambda_x |V_x\rangle\langle V_x|$$

with real eigenvalues.

Let O_y denote the distinct eigenvalues for $y \in \{0, \dots, N' - 1\}$ and

$$\Pi_y = \sum_{x:\lambda_x=O_y} |V_x\rangle\langle V_x|$$

Examples:

1. Z -observable

$$\begin{aligned} Z &= |0\rangle\langle 0| - |1\rangle\langle 1|, \quad O_0 = +1, O_1 = -1, |\Psi\rangle = \alpha|0\rangle + \beta|1\rangle \\ \langle Z \rangle_{|\Psi\rangle} &= (+1)|\langle 0|\Psi\rangle|^2 + (-1)|\langle 1|\Psi\rangle|^2 = |\alpha|^2 - |\beta|^2 \end{aligned}$$

2. X -observable

$$\begin{aligned} X &= |+\rangle\langle +| - |-\rangle\langle -| \\ \langle X \rangle_{|\Psi\rangle} &= |\langle +|\Psi\rangle|^2 - |\langle -|\Psi\rangle|^2 = 2\Re\{\alpha\beta^*\} \end{aligned}$$

3. Multi-qubit local observables: Pauli string

$$P = P_{n-1} \otimes P_{n-2} \otimes \dots \otimes P_0, \quad P_i \in \{I, X, Y, Z\}$$

Let $|\Psi\rangle = |\Psi_{n-1}\rangle \otimes \dots \otimes |\Psi_0\rangle$:

$$\langle P \rangle_{|\Psi\rangle} = \langle \Psi_{n-1} | P_{n-1} | \Psi_{n-1} \rangle \dots \langle \Psi_0 | P_0 | \Psi_0 \rangle$$

Going back to optimization:

The classical minimization problem is hard because it's discrete. We relax it to get:

$$\min g(z), q \in [-1, 1]^n$$

Typically, this does not result in an exact result.

The equivalent quantum problem is:

$$\min \langle \Psi | H | \Psi \rangle \quad |\Psi\rangle \in \mathbb{C}^{2^n}$$

but the only requirement is that $\|\Psi\| = 1$. This however, will result in an exact match between the non-relaxed problem (variational principle).

3.2.1 Variational Principle

Lemma: A computational basis $|x\rangle \in \{|0\rangle, \dots, |N-1\rangle\}$ is an eigenvector of Z_i and Z_{ij} with eigenvalue either $+1$ or -1 .

Proof:

$$\begin{aligned} I &= |0\rangle\langle 0| + |1\rangle\langle 1| \\ Z &= |0\rangle\langle 0| - |1\rangle\langle 1| \end{aligned}$$

By the eigendecomposition property of tensor product. The eigendecomposition of the tensor product is the tensor product of the eigenvectors.

Therefore, for any two different basis vectors $|x\rangle$ and $|y\rangle$:

$$\begin{aligned} \langle y|Z_i|x\rangle &= 0 \\ \langle y|Z_{ij}|x\rangle &= 0 \end{aligned}$$

because $Z_j|x\rangle = Z_{ij}|x\rangle = \pm x$.

Theorem: Discrete and continuous optimization are equivalent!

Discrete:

$$\min \langle x|H|x\rangle, |x\rangle \text{ is computational basis}$$

Continuous

$$\min \langle \Psi|H|\Psi\rangle, |\Psi\rangle \text{ is any state}$$

Proof:

Let:

$$|\Psi\rangle = \sum_{x=0}^{N-1} a_x |x\rangle$$

then

$$\langle \Psi|Z_{ij}|\Psi\rangle = \left(\sum_y a_y^* \langle y|\right) Z_{ij} \left(\sum_x a_x |x\rangle\right) = \sum_y \sum_x a_y^* a_x \langle y|Z_{ij}|x\rangle = \sum_x |a_x|^2 \langle x|Z_{ij}|x\rangle$$

Note that this works because:

$$\langle y|Z_{ij}|x\rangle \text{ is } 0 \text{ if } x \neq y$$

Similarly:

$$\langle \Psi|Z_i|\Psi\rangle = \sum_x |a_x|^2 \langle x|Z_i|x\rangle$$

Therefore:

$$\langle \Psi|H|\Psi\rangle = \sum_{x=0}^{N-1} |a_x|^2 \langle x|H|x\rangle \geq \langle x^*|H|x^*\rangle$$

where $|x^*\rangle$ is the optimal solution to discrete problem.

$\langle x|H|x\rangle \geq \langle x^*|H|x^*\rangle$ and $|a_x|^2 = 1$ by requirement, so the sum above is bounded properly!

Adiabatic QC

- The physical quantum state evolves continuously according to time-dependant Schrodinger equation:

$$H(t)|\Psi_t\rangle = i\hbar \frac{\partial}{\partial t} |\Psi_t\rangle$$

$H(t)$ is the time-dependant Hamiltonian

- basic idea: start with a simple Hamiltonian for which the ground state can be easily prepared.
- slowly change $H(t)$ in such a way that the state $|\Psi_t\rangle$ remains at the ground state at all states.
- The end Hamiltonian is H_1 corresponding to our problem - the final state is the solution!
- DWave machines!!!

Example:

- total time: T
- initial Hamiltonian: H_0
- end Hamiltonian: H_1

Then:

$$\begin{aligned} \implies H(t) &= A(t)H_0 + B(t)H_1 \\ A(0) &= 1, A(T) = 0 \\ B(0) &= 0, B(T) = 1 \\ \text{e.g. } \begin{cases} A(t) = 1 - t/T \\ B(t) = t/T \end{cases} \end{aligned}$$

Quantum Annealing

Now we just have to see how to construct the initial state:

Initial Hamiltonian:

$$H_0 = - \sum_{i=0}^{n-1} X_i$$

where $X_i = I \otimes I \otimes \dots \otimes X \otimes \dots \otimes I$ (where X is the ith element)

Theorem: $|\Psi_0\rangle = |+\rangle^{\otimes n}$ is a ground state of H_0 .

Proof:

$$\begin{aligned}
X &= |+\rangle\langle+| - |-\rangle\langle-| \\
I &= |+\rangle\langle+| + |-\rangle\langle-| \\
|v_x\rangle &\in \{|+\rangle, |-\rangle\}^{\otimes n} \quad \text{form a (Hadamard) basis of } n\text{-qubits} \\
|\Psi\rangle &= \sum_{x=0}^{2^n-1} a_x |v_x\rangle
\end{aligned}$$

This is because you can form any state as a linear combination of a basis!

Then:

$$\begin{aligned}
&\langle\Psi|X_i|\Psi\rangle \\
&= \left(\sum_{y=0}^{2^n-1} a_y^* \langle v_y| \right) X_i \left(\sum_{x=0}^{2^n-1} a_x |v_x\rangle \right) \\
&= \sum_{x=0}^{2^n-1} |a_x|^2 \langle v_x|x_i|v_x\rangle
\end{aligned}$$

and

$$\langle v_x|x_i|v_x\rangle = \begin{cases} 1 & \text{if } v_x^{(i)} = |+\rangle \\ -1 & \text{if } v_x^{(i)} = |-\rangle \end{cases}$$

therefore

$$\sum_{x=0}^{2^n-1} |a_x|^2 \langle v_x|x_i|v_x\rangle \leq 1.$$

But by definition:

$$\langle\Psi_0|X_i|\Psi_0\rangle = 1.$$

Therefore, using the original hamiltonian:

$$\begin{aligned}
H_0 &= - \sum_{i=0}^{n-1} X_i \\
\Rightarrow \langle\Psi|H_0|\Psi\rangle &= - \sum_{i=0}^{n-1} \langle\Psi|X_i|\Psi\rangle \geq -n
\end{aligned}$$

But $\langle\Psi|X_i|\Psi\rangle \geq -1 \Rightarrow |\Psi_0\rangle$ is a ground state of H_0 .

The Quantum Annealing algorithm:

$$\begin{aligned}
H_0 &= \sum_{i=0}^{n-1} -X_i \\
|\Psi_0\rangle &= |+\rangle^{\otimes n} \\
H_1 &= -\sum_{i \neq j} J_{ij} Z_{ij} - \sum_i h_i Z_i \\
H(t) &= (1 - t/T)H_0 + (t/T)H_1, t \in [0, T]
\end{aligned}$$

where T is the annealing time.

$|\Psi_T\rangle$ is the solution to the optimization problem!

3.3 Quantum Approximate Optimization Algorithm (QAOA)

3.3.1 Discretized Adiabatic QC

$$\begin{aligned}
H(t)|\Psi_t\rangle &= i\hbar \frac{\partial}{\partial t} |\Psi_t\rangle \\
H(t) &= A(t)H_0 + B(t)H_1
\end{aligned}$$

Apply discretization:

$$\begin{aligned}
&\begin{cases} tk = k\Delta t \\ \Delta t = \frac{T}{p} \end{cases} \\
\implies H(t_k) &= A(t_k)H_0 + B(t_k)H_1 \\
|\Psi_k\rangle &\approx e^{j\Delta t H(t_k)} |\Psi_{k-1}\rangle \\
\implies |\Psi_p\rangle &\approx \left(\prod_{k=1}^p e^{j\Delta t (A(t_k)H_0 + B(t_k)H_1)} \right) |\Psi_0\rangle \\
\implies |\Psi_p\rangle &\approx \left(\prod_{k=1}^p e^{j\Delta t (A(t_k)H_0)} e^{j\Delta t (B(t_k)H_1)} \right) |\Psi_0\rangle
\end{aligned}$$

when Δt is small! (because the identity $e^{x+y} = e^x e^y$ is not always true for exponentiation of matrices).

Let $\Delta t A(t_k) = \beta_k$ and $\Delta t B(t_k) = \gamma_k$, giving the final approximation:

$$|\Psi_p\rangle \approx \left(\prod_{k=1}^p e^{j\beta_k H_0} e^{j\gamma_k H_1} \right) |\Psi_0\rangle \equiv |\beta\gamma\rangle$$

where $\beta = [\beta_1 \dots \beta_p] \in \mathbb{R}^p$ and $\gamma = [\gamma_1 \dots \gamma_p] \in \mathbb{R}^p$

This is now in terms of circuits!

$$\min_{|\Psi_p\rangle} \langle \Psi_p | H_1 | \Psi_p \rangle = |\beta\gamma\rangle$$

3.3.2 QAOA - a hybrid algorithm

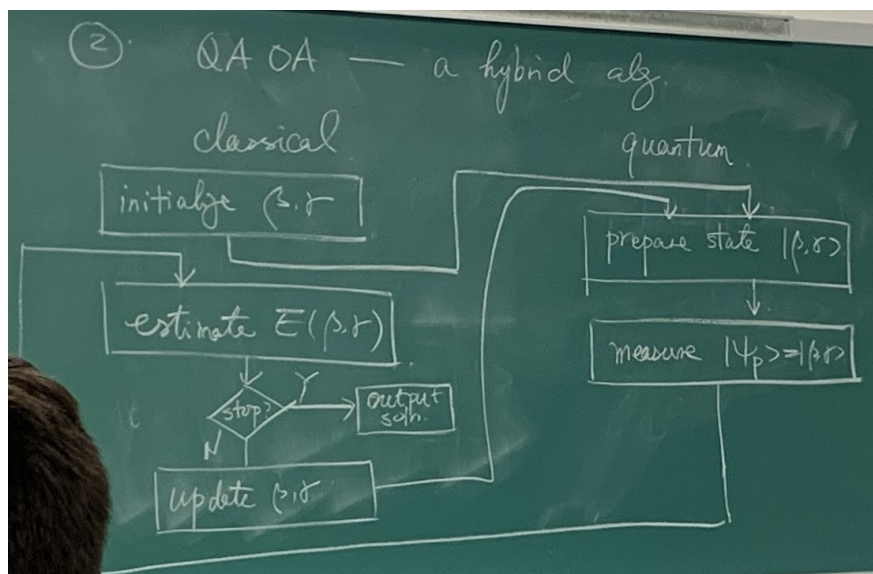


Figure 3.3: QAOA

3.3.3 QAOA Circuit

$$H_0 = - \sum_{i=0}^{n-1} X_i$$

$$H_1 = - \sum_{i \neq j} J_{ij} Z_{ij} - \sum_i h_i Z_i$$

$$|\Psi_0\rangle = |+\rangle^{\otimes n} = H^{\otimes n} |0\rangle$$

Also:

$$e^{j\beta_k H_0} = e^{-j\beta_k \sum_{i=0}^{n-1} X_i}$$

$$= \prod_{i=0}^{n-1} e^{-j\beta_k X_i} \quad \text{because } X_i \text{'s commute (pauli strings)}$$

Each term is just applying $R_x(2\beta_k)$ to the i -th qubit! So essentially applying $R_x(2\beta_k)$ on every qubit!

On the commuting of Pauli strings, we can check:

$$X_1 X_2 = X_2 X_1$$

What about H_1 ?

$$e^{j\gamma_k H_1} = \prod_{i \neq j} e^{-j\gamma_k J_{ij} Z_{ij}} \prod_i e^{-j\gamma_k h_i Z_i} \quad \text{because the } Z\text{'s commute}$$

The second part $e^{-j\gamma_k h_i Z_i}$ is easy because you just apply $R_Z(2\gamma_k h_k)$ to each qubit.

The first part is a little more tricky! Recall:

$$\begin{aligned} Z &= |0\rangle\langle 0| - |1\rangle\langle 1| \\ \Rightarrow Z \otimes Z &= |00\rangle\langle 00| + |11\rangle\langle 11| - |01\rangle\langle 01| - |10\rangle\langle 10| \end{aligned}$$

Recall that the function of a matrix:

$$A = \sum \lambda_k |\lambda_k\rangle\langle \lambda_k| \Rightarrow f(A) = \sum f(\lambda_k) |\lambda_k\rangle\langle \lambda_k|$$

Therefore:

$$\begin{aligned} &\exp\left(-j\frac{\theta}{2} Z \otimes Z\right) \\ &= e^{-j\frac{\theta}{2}}(|00\rangle\langle 00| + |11\rangle\langle 11|) + e^{j\frac{\theta}{2}}(|10\rangle\langle 10| + |01\rangle\langle 01|) \end{aligned}$$

This means that if the qubits have even parity you apply a $-\theta/2$ phase and if odd parity you apply $\theta/2$ phase!

$$|x_1 x_0\rangle = e^{(2y-1)\frac{\theta}{2}} |x_1 x_0\rangle, \quad y = x_1 \oplus x_0, \quad x_i \in \{0, 1\}$$

Recall:

$$R_z(\theta)|y\rangle = e^{(2y-1)\frac{\theta}{2}} |y\rangle$$

3.3.4 Energy Estimation (given circuit)

$$E(\beta, \gamma) = \langle \beta\gamma | H_1 | \beta\gamma \rangle = \langle \Psi_p | H_1 | \Psi_p \rangle$$

$$H_1 = - \sum_{i \neq j} J_{ij} Z_{ij} - \sum_i h_i Z_i$$

Recall: any computational basis state $|x\rangle$ is an eigenvector of Z_i and Z_{ij} with eigenvalue ± 1 .

$$\langle x|Z_i|x\rangle = \begin{cases} 1 & \text{if } |x_i\rangle = |0\rangle \\ -1 & \text{if } |x_i\rangle = |1\rangle \end{cases}, \quad \langle x|Z_{ij}|x\rangle = \begin{cases} 1 & \text{if } |x_i\rangle = |x_j\rangle \\ -1 & \text{if } |x_i\rangle \neq |x_j\rangle \end{cases}$$

Example:

$$\begin{aligned} H_1 &= 3Z_{02} - Z_{12} + 2Z_0 \\ |x\rangle &= |101\rangle \\ \Rightarrow \langle x|H_1|x\rangle &= 3\langle x|Z_{02}|x\rangle - \langle x|Z_{12}|x\rangle + 2\langle x|Z_0|x\rangle = 3(1) - (-1) + 2(-1) = 2. \end{aligned}$$

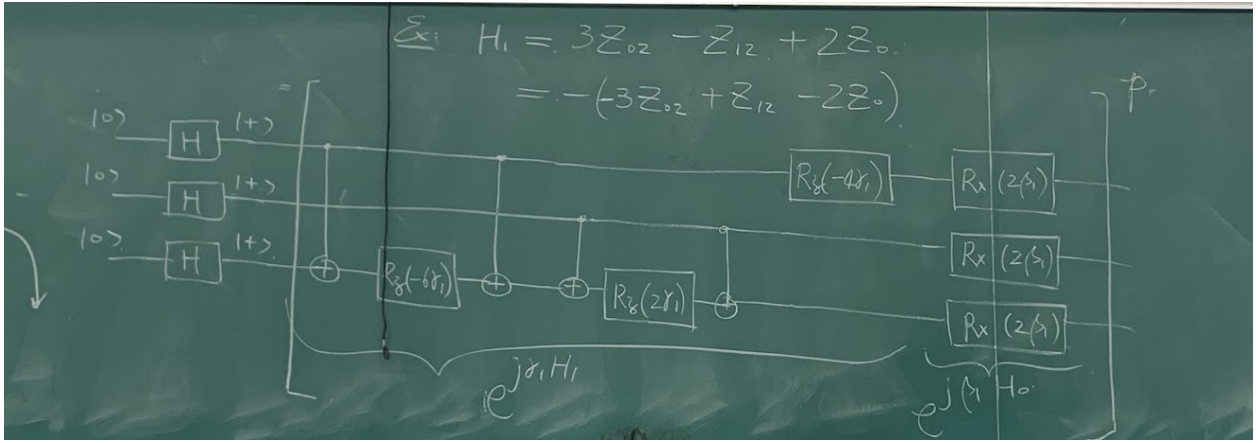


Figure 3.4: QAOA Example

Prepare M copies of the state $|\Psi_p\rangle = |\beta\gamma\rangle$ and measure them, then:

$$E(\beta, \gamma) = \langle \Psi_p | H_1 | \Psi_p \rangle \approx \sum_x \frac{m_x}{M} \langle x | H_1 | x \rangle$$

3.3.5 Parameter training based on gradient descent

$$\min V(\theta) = \langle H \rangle_{|\Psi_\theta\rangle} = \langle \Psi_\theta | H | \Psi_\theta \rangle$$

Vanilla gradient descent: $\theta \leftarrow \theta - \gamma \nabla_\theta V(\theta)$ where γ is a step-size param.

SGD: replace $\nabla_\theta V(\theta)$ by its sample estimate using N_s shots:

$$\theta = \begin{bmatrix} \theta_0 \\ \vdots \\ \theta_{m-1} \end{bmatrix} \Rightarrow \nabla_\theta V(\theta) = \begin{bmatrix} \frac{\partial V(\theta)}{\partial \theta_0} \\ \vdots \\ \frac{\partial V(\theta)}{\partial \theta_{m-1}} \end{bmatrix}$$

e.g.

$$\theta = \begin{bmatrix} \beta_1 \\ \gamma_1 \\ \vdots \\ \beta_p \\ \gamma_p \end{bmatrix}$$

in QAOA.

We cannot use backpropagation to update the values because we cannot observe the values.

1. finite-difference gradient

$$\frac{\partial V(\theta)}{\partial \theta_i} \approx \frac{V(\theta + \epsilon e_i) - V(\theta - \epsilon e_i)}{2\epsilon}$$

where $\epsilon > 0$ is a small number and e_i is the i th unit basis vector.

We need to make $2N_s \times m$ measurements are needed to estimate $\nabla V(\theta)$.

2. simultaneous perturbation stochastic approximation (SPSA)

Let Δ be a random m -dim perturbation vector e.g. $\Delta \in \{+1, -1\}^m$:

$$[\nabla V(\theta)]_i = \frac{V(\theta + \epsilon \Delta) - V(\theta - \epsilon \Delta)}{2\epsilon \Delta_i}$$

This only needs $2N_s$ measurements, independent of m (but more noisy).

3. Quantum Natural Gradient - second order method

$$\theta \leftarrow \theta - \gamma g^{-1}(\theta) \nabla_{\theta} V(\theta)$$

g^{-1} is the quantum Fisher information matrix. This is much more costly in terms of measurement

Because we optimize over the parameters, they are parametric quantum circuits (PQC).

3.3.6 Warm-starting QAOA

- Relax the original QUBO such that $x_i \in [0, 1]$ instead of $x_i \in \{0, 1\}$.
- Solve the relaxed problem using a classical continuous optimizer - semidefinite program (SDP), denote the solution as $\hat{x}_i \in [0, 1]$
- Set H_0 corresponding to $\{\hat{x}_i\}$.
- Let $\theta_i = 2 \sin^{-1}(\hat{x}_i)$

- Then:

$$H_0 = - \sum_{i=0}^{n-1} U_i$$

$$U_i = I \otimes \dots \otimes I \otimes M_i \otimes I \otimes \dots \otimes I$$

$$M_i = R_Y(\theta_i)|0\rangle\langle 0|R_Y(-\theta_i) - R_Y(\theta_i)|1\rangle\langle 1|R_Y(-\theta_i)$$

M_i has the following eigenvalue-eigenvector pairs:

eigenvalue	eigenvector
+1	$R_Y(\theta_i)$
-1	$R_Y(\theta_i)$

The eigenvalues are real, so it is Hermitian and thus an observable!

Therefore, the ground state of H_0 is:

$$R_Y(\theta_{n-1})|0\rangle \otimes R_Y(\theta_{n-2})|0\rangle \otimes \dots \otimes R_Y(\theta_0)|0\rangle$$

Where each θ_i corresponds to $\hat{x}_i$.

Now, we need to change the circuit initialization and the second component of the QAOA circuit.

$$\begin{aligned} e^{j\beta_k M_i} &= e^{j\beta_k R_Y(\theta_i)|0\rangle\langle 0|R_Y(-\theta_i) - e^{-j\beta_k R_Y(\theta_i)|1\rangle\langle 1|R_Y(-\theta_i)} \\ &= R_Y(\theta_i)[e^{j\beta_k}|0\rangle\langle 0| + e^{-j\beta_k}|1\rangle\langle 1|]R_Y(-\theta_i) \\ &= R_Y(\theta_i)R_z(2\beta_k)R_Y(-\theta_i) \end{aligned}$$

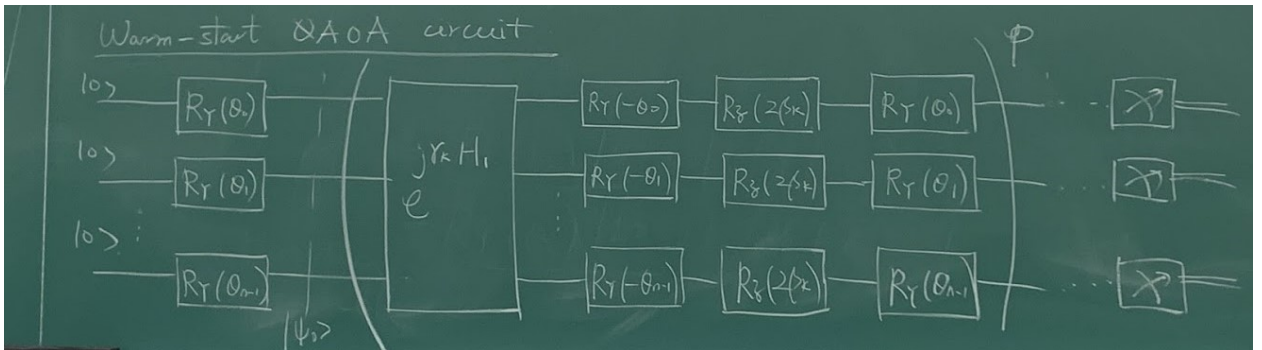


Figure 3.5: Warm Start QAOA

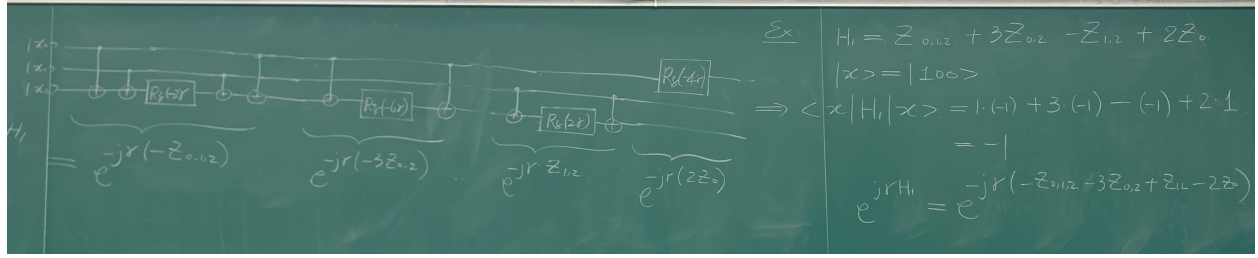


Figure 3.6: Warm Start QAOA Example

3.3.7 High-order binary optimization (HOBO)

- Ising model has terms: $z_{i1}, z_{i2}, \dots, z_{im}$ with $m \geq 2$.
- Hamiltonian: $Z_{i1,i2,\dots,im}$,
- Variational Principle still holds, so min-energy state is a basis state, so QA is still applicable.

QAOA circuit of HOBO:

$$\exp\left(-j\frac{\theta}{2}Z^{\otimes m}\right)|x_{m-1}, \dots, x_0\rangle = e^{j(2y-1)\frac{\theta}{2}}|x_{m-1}, \dots, x_0\rangle$$

where $y = \bigoplus_{i=0}^{m-1} x_i$

Therefore:

$$\begin{aligned} &|x_{m-1}, \dots, x_0\rangle \\ &\rightarrow |y, x_{m-2}, \dots, x_0\rangle \\ &\rightarrow R_z(\theta)|y\rangle \otimes |x_{m-2}, \dots, x_0\rangle \\ &\rightarrow e^{j(2y-1)\frac{\theta}{2}}|x_{m-1}, \dots, x_0\rangle. \end{aligned}$$

Energy estimation: let $|x\rangle$ be in the computational basis

$$\begin{aligned} &\langle x|Z_{i1,i2,\dots,im}|x\rangle \\ &= \begin{cases} 1 & \text{if } x_{i1} \oplus x_{i2} \oplus \dots \oplus x_{im} = 0 \\ -1 & \text{otherwise.} \end{cases} \end{aligned}$$

Satisfiability (SAT) Problem

Given a Boolean expression, determine if there exists $x_i \in \{\text{true}, \text{false}\}$ such that the expression is true.

3.4 Variational Quantum Eigensolver (VQE)

3.4.1 VQE

- Find a ground state of a given Hamiltonian H_1 that describes the energy of a physical or chemical process:

$$\min_{|\Psi\rangle} \langle \Psi | H_1 | \Psi \rangle$$

- H_1 is a linear combination of tensor product of Pauli matrices

$$H_1 = \sum_l a_l G_l$$

$$G_l = \bigotimes_k U_{l,k}, \quad U_{l,k} \in \{X, Y, Z, I\}$$

Therefore, the Ising model and HOBQ are special cases!

- parameterized quantum state:

$$|\Psi(\theta)\rangle = V(\theta)|0\rangle$$

e.g. $V(\theta)$ could be QAOA, variational form (ansatz)

- hardware efficient ansatz (Efficient SU2)
 - alternating layers of R_Y , R_Z and CNOT gates.

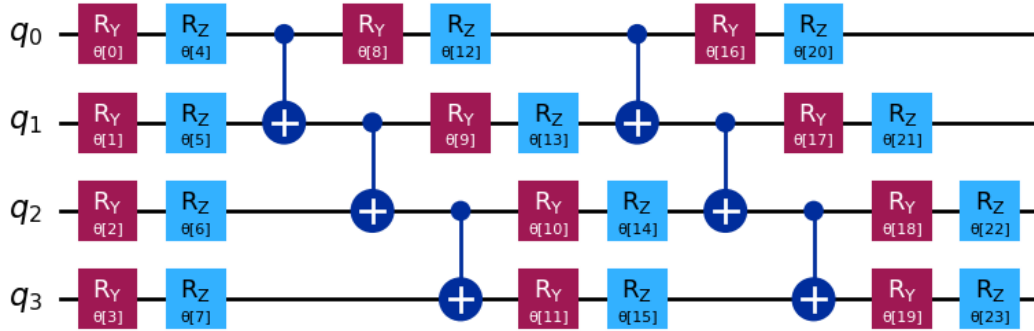


Figure 3.7: Efficient SU2

– $2n \cdot p$ parameters!

- Running the Variational Quantum Eigensolver:
- QAOA is a special case of VQE

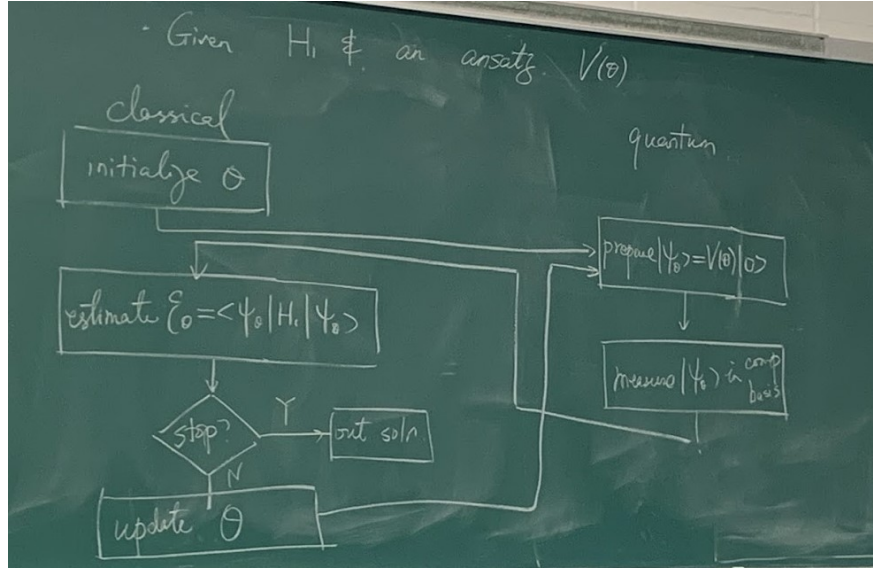


Figure 3.8: VQE

$$\theta = (\beta, \gamma)$$

$$V(\theta) = \prod_{k=1}^p e^{j\beta_k H_0} e^{j\gamma_k H_1}$$

where H_1 is a linear combination of I and Z .

- Analytical gradient: (true for hardware-efficient circuit but not QAOA)

$$\frac{\partial V(\theta)}{\partial \theta_i} = \frac{V(\theta + \frac{\pi}{2} e_i) - V(\theta - \frac{\pi}{2} e_i)}{2}$$

3.4.2 Estimating energy in VQE

For a general observable A :

$$\text{expected value : } \langle A \rangle_{|\Psi\rangle} = \langle \Psi | A | \Psi \rangle$$

$$\text{eigendecomposition : } A = \sum_k \lambda_k |\lambda_k\rangle \langle \lambda_k| = \sum_l \lambda_l |\langle \lambda_l | \Psi \rangle|^2$$

By definition of A being observable, λ_l is real.

Recall:

$$H_1 = \sum_l a_l G_l \implies \langle H_1 \rangle_{|\Psi\rangle} = \sum_l a_l \langle G_l \rangle_{|\Psi\rangle}$$

Therefore, we need to estimate $\langle G_l \rangle_{|\Psi\rangle}$ in the computational basis. We know that G_l is the tensor product of $\{I, X, Y, Z\}$, so the eigenvalues of $G_l \in \{+1, -1\}$ (with eigenvector as the tensor product of the corresponding eigenvectors).

Changes of basis matrices: The eigenbasis of

- Z is: $B_Z = [|0\rangle, |1\rangle] = I$
- X is: $B_x = [|+\rangle, |-\rangle] = HB_z = H$
- Y is: $B_y = [|+j\rangle, |-j\rangle] = SHB_z = SH$

where $S = \begin{bmatrix} 1 & 0 \\ 0 & j \end{bmatrix}$.

Therefore any eigenvector $|\lambda\rangle$ of G can be written as $|\lambda\rangle = U_G|x\rangle$. where U_G is the tensor product of I , H , and SH .

$$|\langle\lambda|\Psi\rangle|^2 = |\langle x|U_G^*|\Psi\rangle|^2$$

Therefore, first apply U_G^* to $|\Psi\rangle$ then measure in computational basis!

For each outcome x , compute the eigenvector $|\lambda_x\rangle = U_G|x\rangle$ and based on this eigenvalue $\lambda_x \in \{+1, -1\}$.

Final estimate:

$$\langle\Psi|G|\Psi\rangle = \frac{m_x}{M}\lambda_x$$

$\underline{\text{Ex:}}$ $G = Z \otimes I \otimes X$
 $\Rightarrow U_G = I \otimes I \otimes H = U_G^*$
 if $|x\rangle = |001\rangle \Rightarrow |\lambda_x\rangle = U_G \cdot |x\rangle$
 $= |00\rangle$
 $\Rightarrow \lambda_x = 1 \cdot 1 \cdot (-1) = -1$

Figure 3.9: Estimating eigenvalue of given eigenvector example

3.5 VQE: find the ground state of a given Hamiltonian

$$H = \sum_l a_l G_l$$

Ground state:

$$|\Psi^*\rangle = \arg \min_{|\Psi\rangle} \langle H \rangle_{|\Psi\rangle} = \arg \min_{|\Psi\rangle} \langle \Psi | H | \Psi \rangle$$

Solution

$$|\Psi^*\rangle = |\lambda_{\min}\rangle$$

The minimum eigenvector of H .

For optimization: QUBO or HOBQ G_l has only I or Z . - QAOA

Then eigenvector of G_l : tensor product of $|0\rangle$ or $|1\rangle$ - computational basis!

$|\Psi^*\rangle$ - groundstate is a computational basis - variational principle!

3.5.1 How to estimate energy based on measuring the state in the computational basis?

Let:

$$|\Psi\rangle = \sum_x b_x |x\rangle \implies \langle H \rangle_{|\Psi\rangle} = \sum_l a_l \langle \Psi | G_l | \Psi \rangle$$

Because $|\Psi\rangle$ is nothing more than an eigenvector of G_l :

$$\langle H \rangle_{|\Psi\rangle} = \sum_l a_l \langle \Psi | G_l | \Psi \rangle = \sum_l a_l \sum_x |b_x|^2 \langle x | G_l | x \rangle$$

The value of $\langle x | G_l | x \rangle$ is just $+1$ or -1 .

The value of $b_x \approx \frac{m_x}{M}$.

3.5.2 General case now

For general VQE where G_l has I, X, Y, Z :

$$G = \sum_{x=0}^{2^n-1} \lambda_x |\lambda_x\rangle \langle \lambda_x|$$

We have $\lambda_x = U_G |x\rangle$.

Example: $G = I \otimes X \otimes Y \otimes Z$

$$\begin{aligned} |\lambda_x\rangle &= |0\rangle \otimes |+\rangle \otimes |-j\rangle \otimes |1\rangle \\ &= (I \otimes H \otimes SH \otimes I)|0011\rangle \end{aligned}$$

And the corresponding eigenvalue is $\lambda_x = 1 \cdot 1 \cdot (-1) \cdot (-1) = 1$

Then, substituting this into the energy equation:

$$\begin{aligned} \langle \Psi | G | \Psi \rangle &= \sum_x \lambda_x |\langle \lambda_x | \Psi \rangle|^2 \\ &= \sum_x \lambda_x |\langle x | U_G^* | \Psi \rangle|^2 \end{aligned}$$

First, we apply U_G^* to $|\Psi\rangle$ then measure in the computational basis.

Then, if we let $|\Psi\rangle = \sum_x c_x |\lambda_x\rangle = \sum_x c_x U_G |x\rangle$, then $U_G^* |\Psi\rangle = \sum_x c_x |x\rangle$.

$$\begin{aligned} \langle H \rangle_{|\Psi\rangle} &= \sum_l a_l \langle \Psi | G_l | \Psi \rangle \\ &= \sum_l a_l \sum_x \lambda_{l,x} |\langle x | U_{G,l}^* | \Psi \rangle|^2 \\ &\approx \sum_l a_l \sum_x \lambda_{l,x} c_{l,x}^2 \\ &\approx \sum_l a_l \sum_x \lambda_{l,x} \frac{m_x}{M} \end{aligned}$$

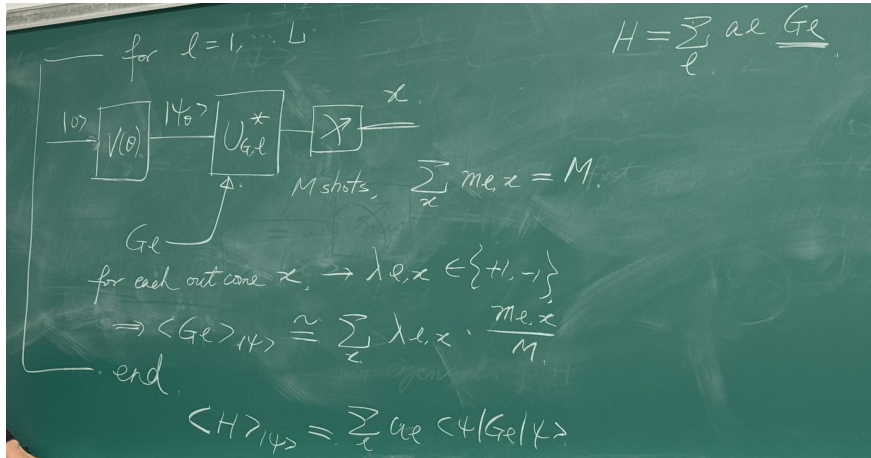


Figure 3.10: VQE Algorithm

3.5.3 Finding excited states

$$\lambda_0 < \lambda_1 < \lambda_2$$

VQE finds $\lambda_0 = \min \langle \Psi | H | \Psi \rangle$

How do you find λ_1 ?

$$H' = H + c|\lambda_0\rangle\langle\lambda_0| \longrightarrow \{|\lambda_j\rangle\} \text{ are still eigenvectors of } H'$$

Then

$$\begin{aligned} H'|\lambda_0\rangle &= \lambda_0|\lambda_0\rangle + c|\lambda_0\rangle = (\lambda_0 + c)|\lambda_0\rangle \\ (j \neq 0) : H'|\lambda_j\rangle &= H|\lambda_j\rangle = \lambda_j|\lambda_j\rangle \end{aligned}$$

If $\lambda_1 < \lambda_0 + c \implies c > \lambda_1 - \lambda_0$, then the ground state of H' is $|\lambda_1\rangle$! by running VQE on H' we obtain λ_1 , can further find λ_2 .

But $|\lambda_0\rangle\langle\lambda_1|$ is not a Pauli string. How do we estimate $|\lambda_0\rangle\langle\lambda_0|_{|\Psi\rangle}$?

Recall:

$$\begin{aligned} |\lambda_0\rangle &= V(\theta_0)|0\rangle - \text{ground state} \\ |\Psi\rangle &= V(\theta)|0\rangle - \text{excited state} \\ \implies |\langle\lambda_0|\Psi\rangle|^2 &= |\langle 0|V(\theta_0)^*V(\theta)|0\rangle|^2 \\ &= P(\text{obtaining } |0\rangle \text{ by measuring } V(\theta_0)^*V(\theta)|0\rangle \text{ in comp. basis}) \end{aligned}$$

Chapter 4

Quantum Machine Learning

Data-focused as opposed to having an analytical objective function! Different types:

- QNN
- Quantum Kernel (using the states)

4.1 Quantum Neural Network

Objective: Binary function to be learned

$$f : \mathbb{R}^n \rightarrow \{0, 1\} \text{ or } \{-1, 1\}$$

training data set:

$$S = \{(x_i, y_i), i = 1, \dots, k\}$$

4.1.1 Classical NN for binary classification

1. Feed-forward NN

[NN review]

4.1.2 Variational Quantum Classifier

Composed of a feature map, ansatz, and projective measurement.

- Feature map: maps $|0\rangle$ states and x data values to some state mapping $\phi(x)$.
- ansatz is the parametric neural network
- the projective measurement converts all n qubits into one measurement.

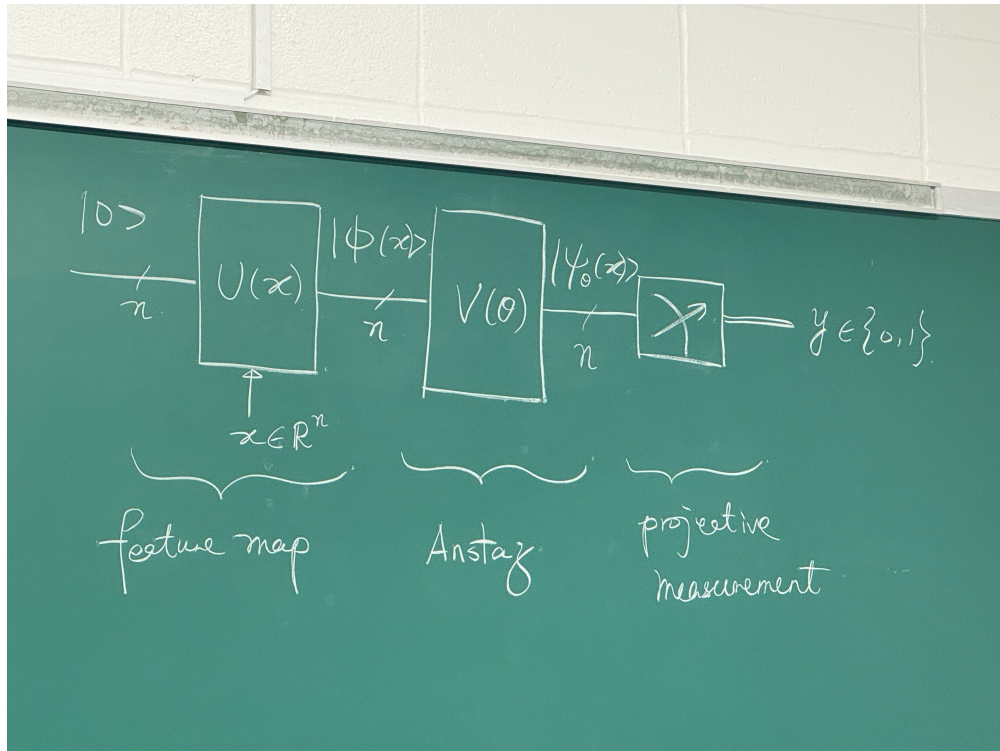


Figure 4.1: QNN

Measurement and loss function

Given an n qubit input $|\Psi_\theta(x)\rangle$ to a measurement circuit, if we expect only one output y , we can only check the partial or parity measurements.

Recall: Cross-entropy loss

$$L(\theta) = \sum_{i=1}^k l_\theta(x_i, y_i)$$

$$l_\theta(x, y) = -\log p_\theta(y|x)$$

QNN can also use linear loss:

$$l_\theta(x, y) = 1 - p_\theta(y|x) \quad (\text{error prob.})$$

Given θ , how to estimate $L(\theta)$?

```

1 for i = 1, ..., K (training sample)
2   - prepare M copies of  $|\Psi_{\theta}(x_i)\rangle$  and measure each to get  $z_i^l$  in  $\{0, 1\}$ 
3   - define sample loss:  $p_{\theta}(y_i|x_i) = 1/M \sum_{l=1}^M 1\{z_i^l = y_i\}$ 
4 end
5 compute the loss using the sums of the sample losses

```

- Total number of shots for one evaluation of $L(\theta)$ is MK as there are K steps and M copies per step.
- Total number of shots for evaluation of $\nabla L(\theta)$:
 - SPSA - $2MK$
 - Param. hisft rule (num. diff) - $2MK \dim(\theta)$

Projective measurement: As we only have two expected outputs, we define $\pi_1 + \pi_2 = I$. We can then define the probabilities of outcomes in terms of projective measurements:

$$\begin{aligned} p(y = 0) &= \langle \Psi | \pi_0 | \Psi \rangle \\ p(y = 1) &= \langle \Psi | \pi_1 | \Psi \rangle \end{aligned}$$

Then, the loss is simply:

$$L(\theta) = - \sum_{i=1}^K \langle \pi_{y_i} \rangle_{|\Psi_\theta(x_i)\rangle}$$

feature map (a.k.a. data encoding/loading)

Classical data $x \in \mathbb{R}^n \longrightarrow$ quantum state $|\Psi(x)\rangle = U(x)|0\rangle$

- Amplitude encoding - maps a 2^n -dim classical unit vector to n qubits.

$$\begin{aligned} x \in \mathbb{R}^{2^n}, \quad \|x\| &= 1 \\ \longrightarrow |\Phi(x)\rangle &= \sum_{i=0}^{2^n-1} x_i |i\rangle \\ U(x) &= \text{diag}(x_0, \dots, x_{2^n-1}) \end{aligned}$$

- advantage: requires n qubits to encode 2^n elements
- disadvantages: subsequent alg. must operate on amplitude, method to prepare state is not efficient
- Angle encoding - map each element x_i to a qubit

$$x \in \mathbb{R}^n \longrightarrow |\Phi(x)\rangle = \bigotimes_{i=0}^{n-1} R_Q(x_i) |0\rangle, Q \in \{X, Y, Z\}$$

For R_X and R_Y this is perfectly fine. If you apply R_Z you have to apply H first to provide your encoding otherwise if you apply R_Z to $|0\rangle$ it doesn't do anything.

- Efficient but it doesn't provide entanglement, which gives it little advantage.
- Pauli feature map (with entanglement)
 - To have an advantage over classical approaches, need a map based on circuits that are hard to simulate classically

- in general, consists of layers of Hadamard gates interleaved with entangling blocks
- with each entangling block are exp. of Pauli strings

Example: ZZ feature map

$$U(x) = \exp \left\{ -j \left(\sum_{i=0}^{n-1} x_i Z_i + \sum_{i < j} (\pi - x_i)(\pi - x_j) Z_{ij} \right) \right\}$$

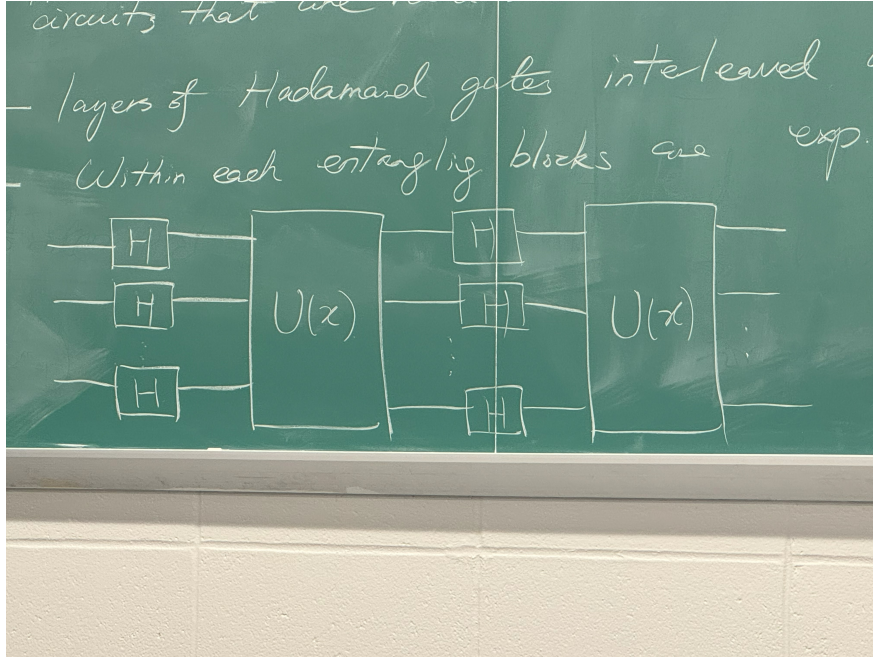


Figure 4.2: Feature mapping

Parameterized quantum circuit (PQC)/Variational form/ansatz

- Mean-field ansatz: use only single-qubit gates - output separable states

$$R(\theta_1, \theta_2, \theta_3) = R_Q(\theta_1) \cdot R_{Q'}(\theta_2) \cdot R_Q(\theta_3), \quad \text{where } Q, Q' \in \{X, Y, Z\}, Q \neq Q'.$$

- Hardware-efficient ansatz - single qubit-gates + fixed entanglement

$$\begin{aligned} V(\theta) &= V_L(\theta) V_{L-1}(\theta) \dots V_1(\theta) \\ V_l(\theta) &= U_{\text{ent}}(R(\theta_{n-1}) \otimes \dots \otimes R(\theta_0)), \quad l = 1, \dots, L \\ U_{\text{ent}} &\in \{CNOT, CZ\} \text{ gates} \end{aligned}$$

Analytical Gradient - parametric shift rule

- Assume each parameter θ_i contributes to the PQC through a gate of the form

$$U_i(\theta) = \exp\left(-j\frac{\theta_i}{2}P_i\right)$$

where P_i is a Pauli string.

- satisfied by all ansatzes we've seen so far besides the QAOA ansatz

$$\frac{\partial V(\theta)}{\partial \theta_i} = \frac{V(\theta + \frac{\pi}{2}e_i) - V(\theta - \frac{\pi}{2}e_i)}{2}$$

Proof: $\theta = \{\theta_1, \dots, \theta_M\}$ for each $k = 1, \dots, M$:

$$U(\theta) = V(\theta_{-k})U_k(\theta_k)W(\theta_{-k})$$

$$\theta_{-k} = \theta \setminus \theta_k$$

Then:

$$\begin{aligned}\langle O \rangle_{|\Psi(\theta)\rangle} &= \langle \Psi(\theta) | O | \Psi(\theta) \rangle \\ &= \langle 0 | U(\theta)^* \cdot O \cdot U(\theta) | 0 \rangle \\ &= \langle 0 | W(\theta_{-k})^* U_k(\theta_k)^* V(\theta_{-k})^* O V(\theta_{-k}) U(\theta_k) W(\theta_{-k}) | 0 \rangle \\ &= \langle \phi_k | U_k(\theta_k)^* R_k U_k(\theta_k) | \phi_k \rangle\end{aligned}$$

and

$$\begin{aligned}\frac{\partial \langle O \rangle_{|\Psi(\theta)\rangle}}{\partial \theta_k} &= \langle \phi_k | \frac{\partial U_k(\theta_k)^*}{\partial \theta_k} R_k U_k(\theta_k) | \phi_k \rangle + \langle \phi_k | U_k(\theta_k)^* R_k \frac{\partial U_k(\theta_k)}{\partial \theta_k} | \phi_k \rangle \\ \frac{\partial U_k(\theta_k)}{\partial \theta_k} &= \frac{\partial}{\partial \theta_k} \exp(-j\frac{\theta_k}{2}P_k) \\ &= -\frac{j}{2} \exp(-j\frac{\theta_k}{2}P_k) \\ \implies \frac{\partial U_k(\theta_k)}{\partial \theta_k} &= -\frac{j}{2} P_k U_k(\theta_k) \implies U_k(\theta_k) + 2 \frac{\partial U_k(\theta_k)}{\partial \theta_k} = (I - jP_k)U_k(\theta_k) = S_k^+\end{aligned}$$

Barren Plateaus

- The loss function $L(\theta)$ exhibits a barren plateau if:

$$\mathbb{E}\{\nabla L(\theta)\} = 0$$

$$\text{Var}\{\nabla L(\theta)\} \sim \frac{1}{b^2}$$

for some $b > 1$ where n is the number of qubits

- By randomly initializing the parameter θ with high prob. one ends up in a flat region of the loss function where the gradient is exponentially suppressed.
- fundamental issue of QNN

4.2 Quantum Support Vector Machines

4.2.1 Classical SVM

Linear binary classification

- binary target function to be learned:

$$f : \mathbb{R}^n \longrightarrow \{-1, +1\}$$

- training data set $S = \{(x_1, y_1), \dots, (x_k, y_k)\}$
- B : is the set of candidate functions - linear classifiers

$$B = \{x \longrightarrow \text{sign}(w \cdot x + b), w \in \mathbb{R}^n, b \in \mathbb{R}\}$$

We know that $wx + b = 0$ is just a hyperplane in $\mathbb{R}^n$:

$$\text{sign}(wx + b) = \begin{cases} +1, & x \text{ is on one side} \\ -1, & x \text{ is on other side} \end{cases}$$

Separable case

- Assume that the training set S is linearly separable: i.e. there exists (w, b) such that:

$$y_i(wx_i + b) \geq 0, i = 1, \dots, k$$

- There are an infinite number of such hyperplanes, so we maximize the minimum distance from the hyperplane. The minimum distance from some point $z \in \mathbb{R}^n$ to the hyperplane is:

$$\rho(z) = \frac{|wz + b|}{\|w\|_2}$$

- The geometric margin of a linear classifier h for a training set S (distance of the hyperplane to the closest sample point).

$$\rho_h = \min\{\rho(x_1), \dots, \rho(x_k)\}$$

- SVM formulation

$$\begin{aligned}
h^* &= \operatorname{argmax}_{h \in B} \rho_h \\
\Rightarrow \rho^* &= \max_{(w,b)} \min_{1 \leq i \leq k} \frac{|wx_i + b|}{\|w\|}, \quad \text{s.t. } y_i(wx_i + b) \geq 0 \\
\{\text{Since } S \text{ is separable}\} &\longrightarrow = \max_{(w,b)} \min_{1 \leq i \leq k} \frac{y_i(wx_i + b)}{\|w\|} \\
\{\text{objective is invariant to scaling}\} &\longrightarrow = \max_{(w,b)} \frac{1}{\|w\|}, \quad \min y_i(wx_i + b) = 1 \\
&= \max_{y_i(wx_i + b) \geq 1} \frac{1}{\|w\|} \\
&\Rightarrow \text{primal formulation:} \\
&\min_{w,b} \frac{1}{2} \|w\|^2 \\
&\text{s.t. } y_i(wx_i + b) \geq 1, i = 1, \dots, k
\end{aligned}$$

Convex programs:

$$\begin{aligned}
&\min_{x \in \mathbb{R}^n} f(x) \\
&\text{s.t. } g_i(x) \leq 0, i = 1, \dots, k
\end{aligned}$$

- Lagrangian:

$$L(x, \alpha) = f(x) + \sum_{i=1}^k \alpha_i g_i(x), \quad \alpha_i \geq 0, i = 1, \dots, k$$

- KKT Conditions: x^* is optimal iff there exists α^* s.t.

$$\begin{cases} \frac{\partial L(x^*, \alpha^*)}{\partial x} = 0 \\ \alpha_i^* g_i(x^*) = 0 \end{cases} \Rightarrow \begin{cases} \alpha_i^* > 0 : \longrightarrow g_i(x^*) = 0 - \text{active constraint} \\ \alpha_i^* = 0 \longrightarrow g_i(x^*) \leq 0 - \text{inactive constraint} \end{cases}, \quad \alpha_i^* \geq 0$$

- Dual problem:

$$\max_{\alpha \geq 0} \Theta(\alpha), \Theta(\alpha) = \min_{x \in \mathbb{R}^d} d(x, \alpha)$$

For convex problems, the optimal solution to the dual is the same as the optimal solution to the primal.

$$\Theta(\alpha^*) = f(x^*)$$

Formulating the problem as a Lagrangian:

$$L(w, b, \alpha) = \frac{1}{2} \|w\|^2 + \sum_{i=1}^k \alpha_i [1 - y_i(wx_i + b)]$$

The KKT conditions:

$$\begin{aligned}\frac{\partial L}{\partial w} &= w - \sum_{i=1}^k \alpha_i y_i x_i = 0 \\ \implies w &= \sum_{i=1}^k \alpha_i y_i x_i\end{aligned}$$

The optimal solution is nothing but a linear combination of the data vectors.

$$\begin{aligned}\frac{\partial L}{\partial b} &= - \sum_{i=1}^k \alpha_i y_i = 0 \\ \implies \sum_{i=1}^k \alpha_i y_i &= 0.\end{aligned}$$

Finally,

$$\alpha_i [1 - y_i (wx_i + b)] = 0, i = 1, \dots, k$$

Vector x_i appears in w iff. $\alpha_i > 0 \leftrightarrow wx_i + b = \pm 1 \implies x_i$ is a support vector.

We only need to find $\{\alpha_i\}$ by solving the dual problem.

$$\begin{aligned}L(w, b, \alpha) &= \frac{1}{2} \|w\|^2 + \sum \alpha_i [1 - y_i (wx_i + b)] \\ &= \frac{1}{2} \|w\|^2 + \sum \alpha_i - \sum \alpha_i y_i wx_i - b \sum \alpha_i y_i \\ \xrightarrow{w = \sum \alpha_i y_i x_i} \Theta(\alpha) &= \sum_i \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j (x_i \cdot x_j)\end{aligned}$$

Therefore, we have the dual problem (quadratic objective + linear constraints):

$$\begin{aligned}\max \quad & \Theta(\alpha) \\ \alpha_i & \geq 0 \\ \sum_{i=1}^k \alpha_i y_i &= 0\end{aligned}$$

For any support vector x_i , $wx_i + b = y_i \implies b = y_i - wx_i$.

Finally, the SVM classifier is given by:

$$\begin{aligned}h(x) &= \text{sign}(wx + b) \\ &= \text{sign} \left(\sum_{i=1}^k \alpha_i y_i (x_i \cdot x) + b \right)\end{aligned}$$

Note that data only appears inside an inner product!

4.2.2 Kernel Method

- if linear separation is not possible
- nonlinear decision boundary:
 1. map the input space x to a higher dim feature space $H : x \rightarrow \phi(x) \in H$
 2. linearly separate $\phi(x)$ in H using SVM.

In feature space, replace (x_i, x) by $K(x_i, x) = (\phi(x_i), \phi(x))$

Kernels:

$$K : X \times X \rightarrow \mathbb{R} \text{ for } x_1, x_2 \in X$$

$$K(x_1, x_2) = (\phi(x_1), \phi(x_2)) \quad \{\text{inner product}\}$$

In classical learning theory, you only have K , not ϕ .

- Polynomial kernel:

$$K(x, z) = (xz + c)^d$$

Example: $n = 2, d = 2$:

$$K(x, z) = (x_1 z_1 + x_2 z_2 + c)^2 = \begin{bmatrix} x_1^2 \\ x_2^2 \\ \sqrt{2}x_1x_2 \\ \sqrt{2}cx_1 \\ \sqrt{2}cx_2 \\ c \end{bmatrix} \cdot \begin{bmatrix} z_1^2 \\ z_2^2 \\ \sqrt{2}z_1z_2 \\ \sqrt{2}cz_1 \\ \sqrt{2}cz_2 \\ c \end{bmatrix}$$

- Gaussian Kernel/RBF

$$K(x, z) = \exp\left(-\frac{\|z - x\|^2}{2\sigma^2}\right)$$

4.2.3 QSVM

Quantum kernel - feature map

- classical data: $x \in \mathbb{R}^n \rightarrow |\phi(x)\rangle \in \mathbb{C}^{2^n}$
- $K(x_i, x_j) = |\langle \phi(x_i) | \phi(x_j) \rangle|^2$

$$|\phi(x)\rangle = U(x) \cdot |0\rangle, \quad \text{where } U(x) \text{ is the feature map parametric circuit}$$

Then:

$$K(x_i, x_j) = |\langle 0 | U^*(x_i) U(x_j) | 0 \rangle|^2$$

Therefore, run the quantum circuit $U^*(x_i)U(x_j)$ on input $|0\rangle$ and measure the probability of 0 output.

Quantum kernel estimation:

```

1  for i=1,...,k do:
2      k(x_i, x_j)=1
3      for j=i+1,...,k do:
4          Q -> apply U*(x_i)U(x_j) on |0>
5          Q -> measure probability of 0 output - p
6          K(x_i, x_j) = K(x_j, x_i)=p
7      end j
8  end i

```

Test stage: given data x

```

1  for i=1,...,k do:
2      Q -> apply U*(x)U(x_i) on |0>
3      Q -> measure probability of 0 output - p
4      K(x, x_i)=p
5  end i

```

Quantum SVM

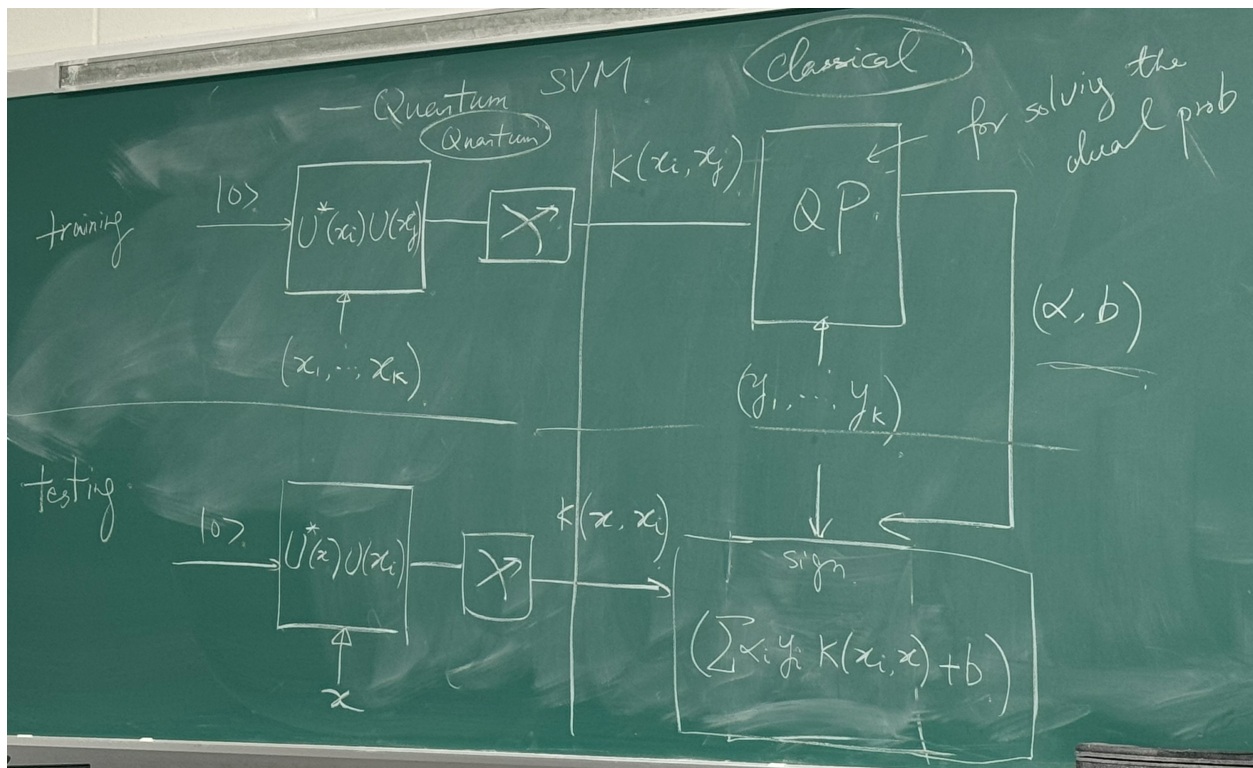


Figure 4.3: qSVM

Differences from classical SVM:

- quantum feature maps are more expressive than classical ones
- quantum kernel estimation is noisy, classical kernel is exact!
- classical kernel bypass the mapping ϕ , quantum explicitly constructs ϕ

4.2.4 Dimension reduction - PCA

Data matrix $X = [x_1, \dots, x_k] \in \mathbb{R}^{n \times k}$

- an orthogonal projector $P = UU^\top \in \mathbb{R}^{n \times n}$ with $U \in \mathbb{R}^{n \times d}$ containing d orthogonal vectors, $d < n$.

For any $x \in \mathbb{R}^n$ $UU^\top x$ projects x onto a d -dimensional subspace.

$$x \approx \sum_{l=1}^d \alpha_l u_l$$

Turns out, the vectors u_l are just the eigenvectors of the correlation matrix of X .

PCA Solution:

- For autocorrelation matrix $C = XX^\top \in \mathbb{R}^{n \times n}$
- Eigendecomposition of matrix: $C = \sum_{i=1}^n \lambda_i u_i u_i^\top$ in the order of largest to smallest eigenvalues.
- Projection matrix $U = [u_1, u_2, \dots, u_d] \in \mathbb{R}^{n \times d}$, largest eigenvectors contain most relevant directions in data